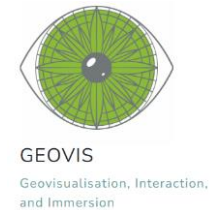




Project Oracles: Visualization of ensemble forecasts of the coastal flooding phenomenon to aid decision support under uncertainty

Julius Bañgate, Jacques Gautier, Sidonie Christophe



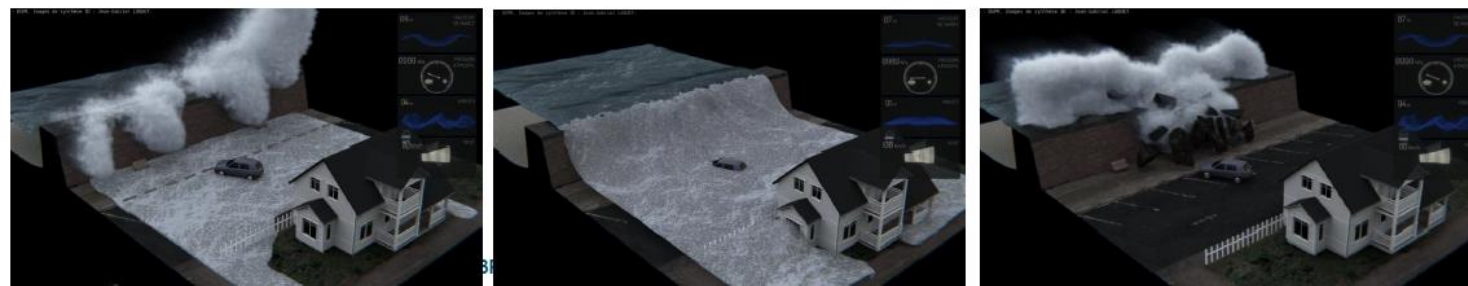
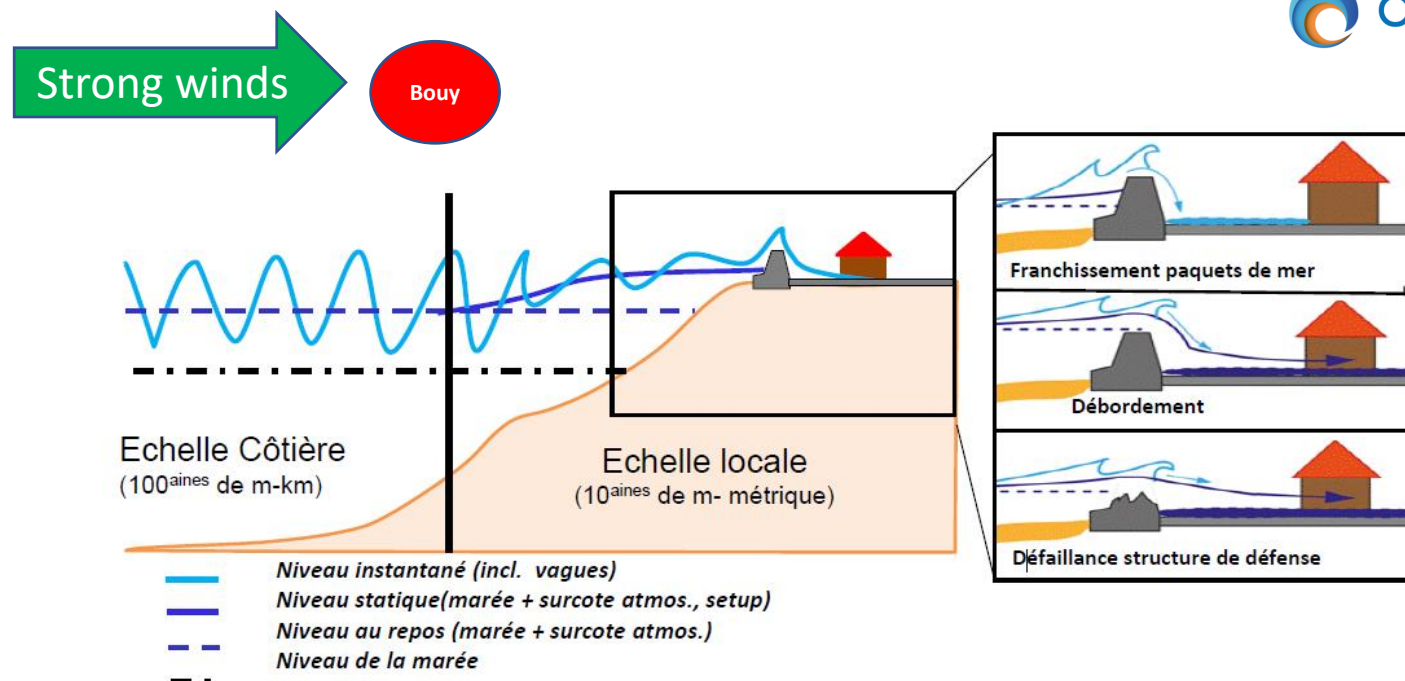
Journée de la recherche 2025

March 11, 2025, Marne-la-Vallée



Objective

“Towards the use of **ensemble forecasts** of marine submersion to aid decision support under **uncertainty**.”





NWP models

Simulation of meteo-oceanic conditions (wind, waves, sea level, surcote, tides)

⇒ **Analysis** of the results by Météo-France forecasters to define a VVS alert level (wave submersion vigilance)

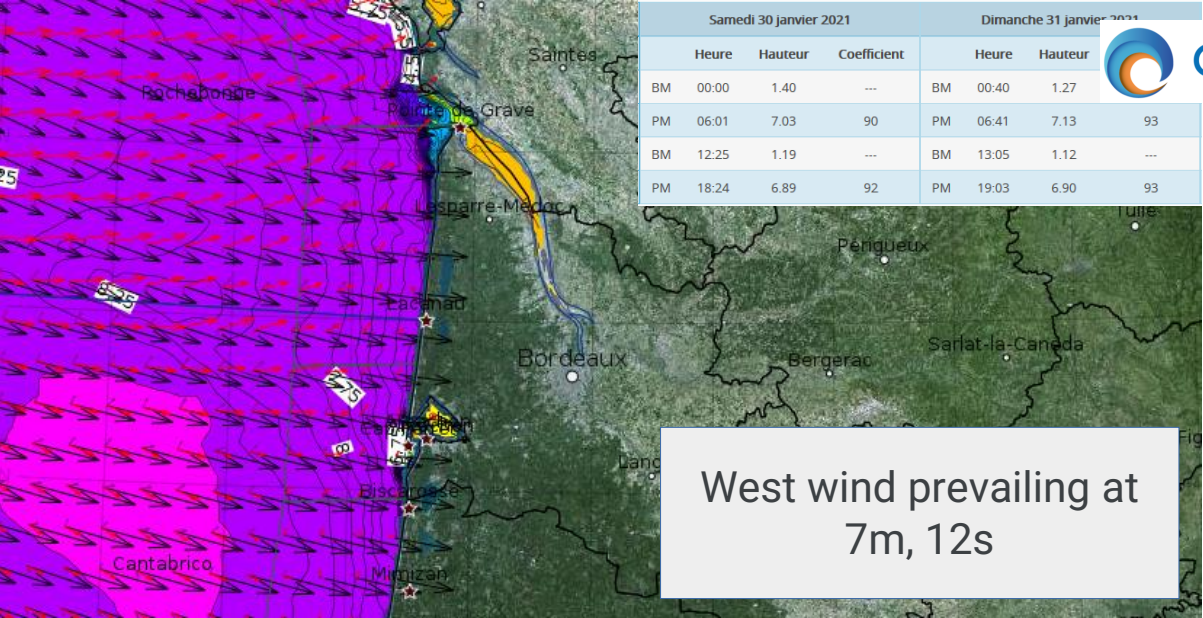
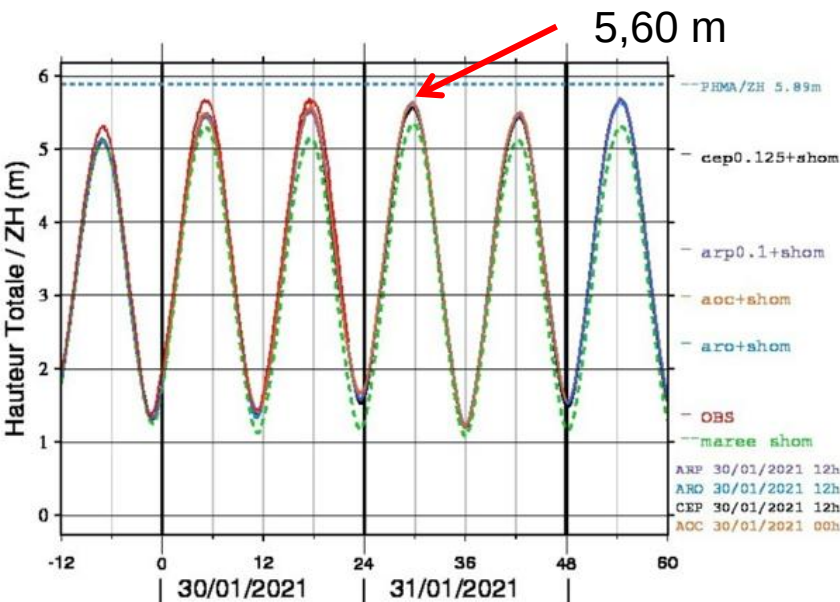


Coastal submersion models

Simulation of marine submersion phenomena (water height on land) based on the results of Météo France models

⇒ **Analysis** of the results and the VVS alert level by **local authorities** (town hall, Direction départementale de territoires et de la mer (DDTM), Prefecture) to define a level of prevention.

Workflow Expertise VVS



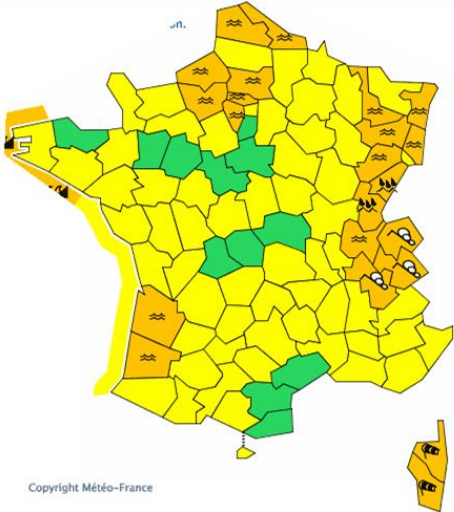
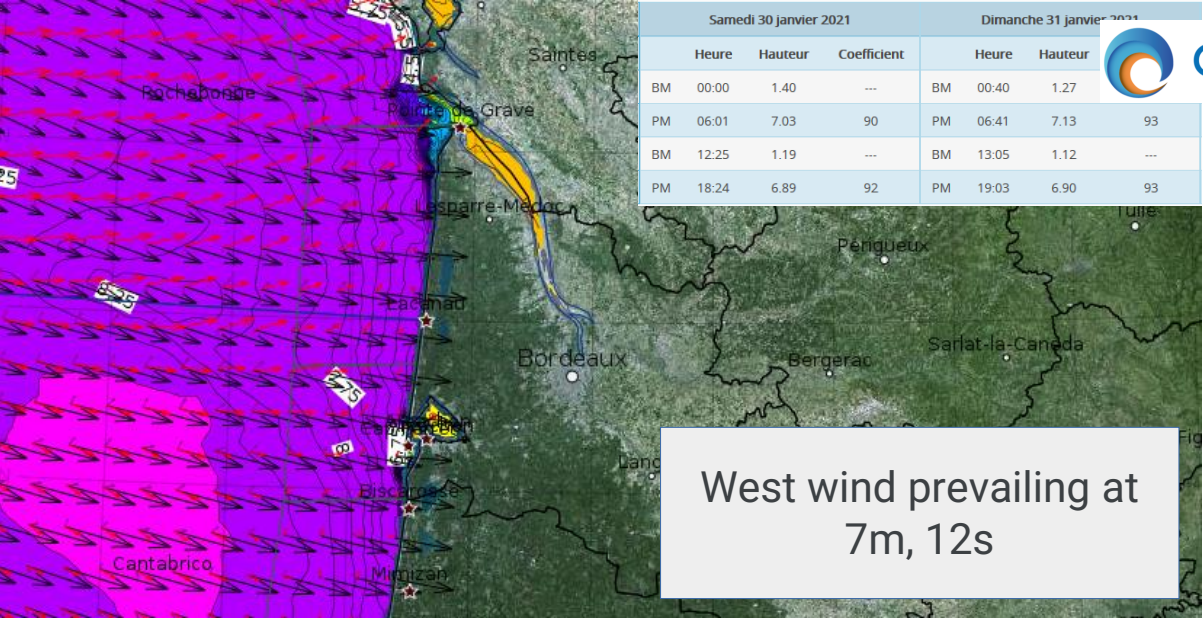
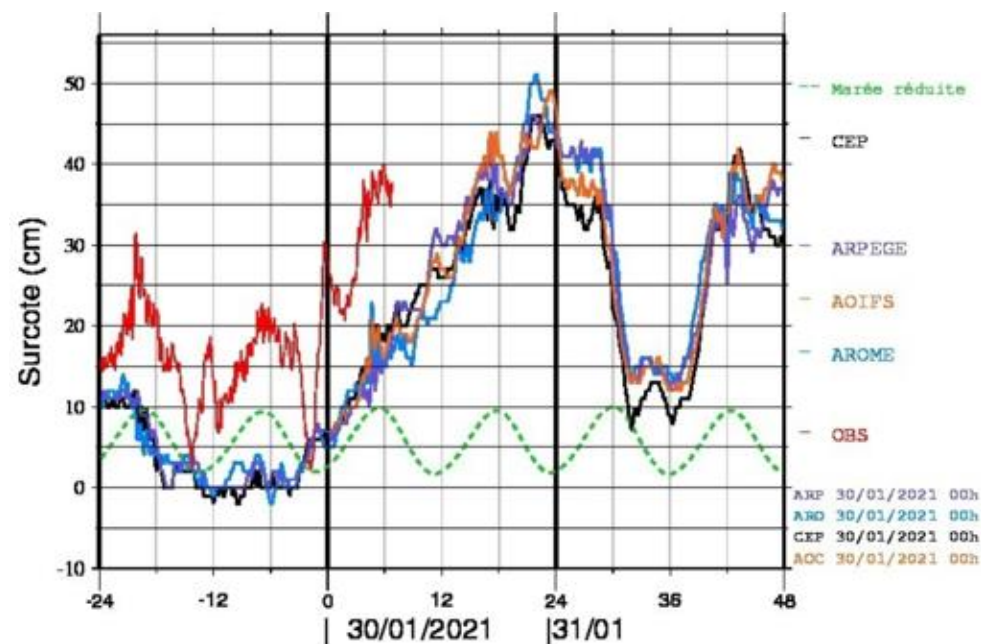
- Une vigilance absolue s'impose des phénomènes dangereux d'intensité exceptionnelle sont prévus...
- Soyez très vigilant , des phénomènes dangereux sont prévus ...
- Soyez attentif si vous pratiquez des activités sensibles au risque météorologique ...
- Pas de vigilance particulière.

Ocean Name

Port A

H eau totale (m) \ H1/3 (m)	[0: h1]	[h1 : h2]	[h2: h3]	[h3: h4]	[h4: h5]	[h5: h6]	[h6: h7]	[h7: h8]
[v4 : v5]	Impossible	Impossible	Impossible	Improbable				
[v3 : v4]	Impossible	Impossible	Improbable					
[v2 : v3]								
[v1 : v2]								
[0 : v1]					P>12			

Workflow Expertise VVS



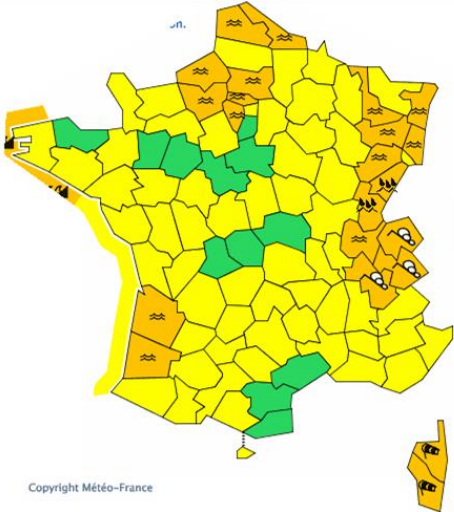
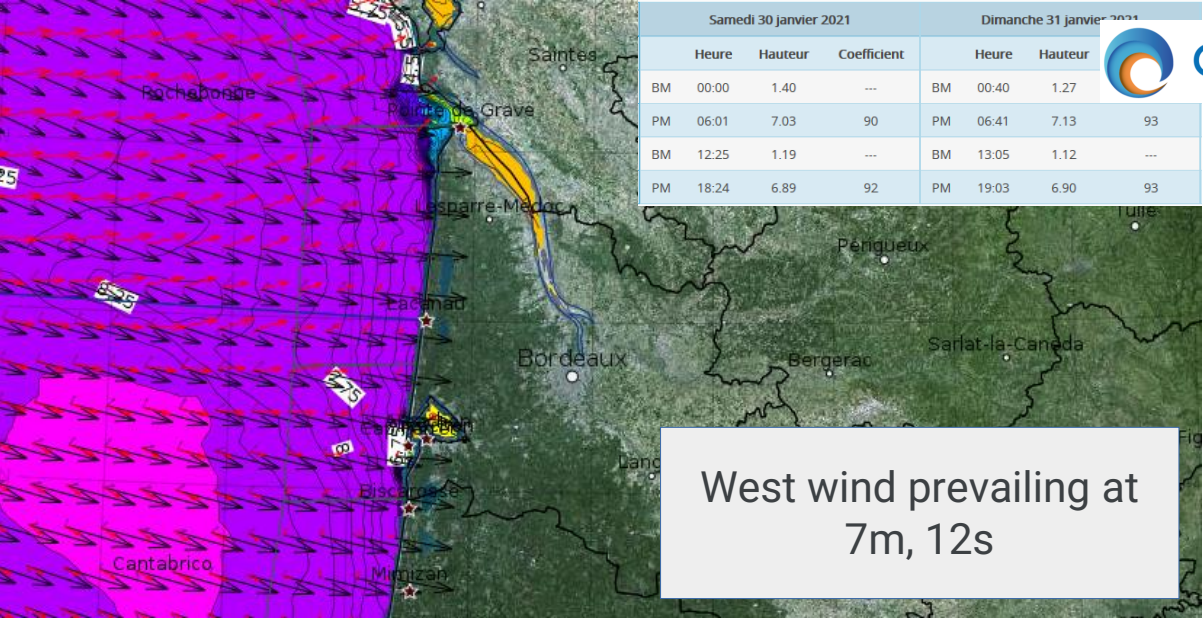
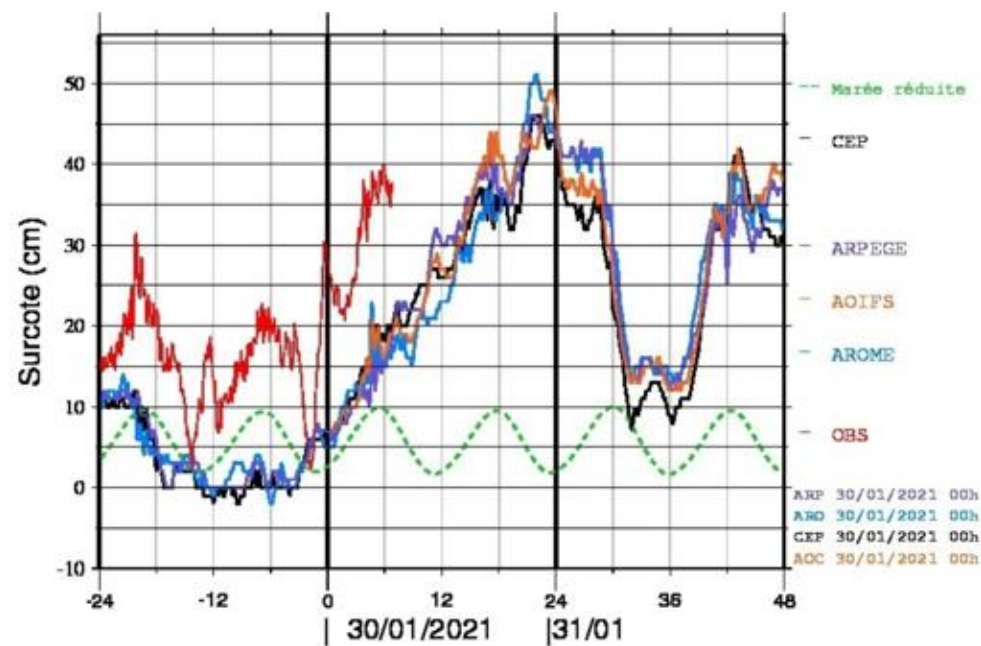
- Une vigilance absolue s'impose des phénomènes dangereux d'intensité exceptionnelle sont prévus...
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- Soyez attentif si vous pratiquez des activités sensibles au risque météorologique ...
- Pas de vigilance particulière.

Ocean Name

Port A

H1/3 (m)	[0: h1]	[h1 : h2]	[h2: h3]	[h3: h4]	[h4: h5]	[h5: h6]	[h6: h7]	[h7: h8]
H eau totale (m)								
[v4 : v5]	Impossible	Impossible	Impossible	Improbable				
[v3 : v4]	Impossible	Impossible	Improbable					
[v2 : v3]								
[v1 : v2]								
[0 : v1]					P>12			

Workflow Expertise VVS



- Une vigilance absolue s'impose** des phénomènes dangereux d'intensité exceptionnelle sont prévus...
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- Pas de vigilance particulière.**

Ocean Name

Port A

H eau totale (m) \ H1/3 (m)								
	[0 : h1]	[h1 : h2]	[h2 : h3]	[h3 : h4]	[h4 : h5]	[h5 : h6]	[h6 : h7]	[h7 : h8]
[v4 : v5]	Impossible	Impossible	Impossible	Improbable				
[v3 : v4]	Impossible	Impossible	Improbable					
[v2 : v3]								
[v1 : v2]								
[0 : v1]					P>12			

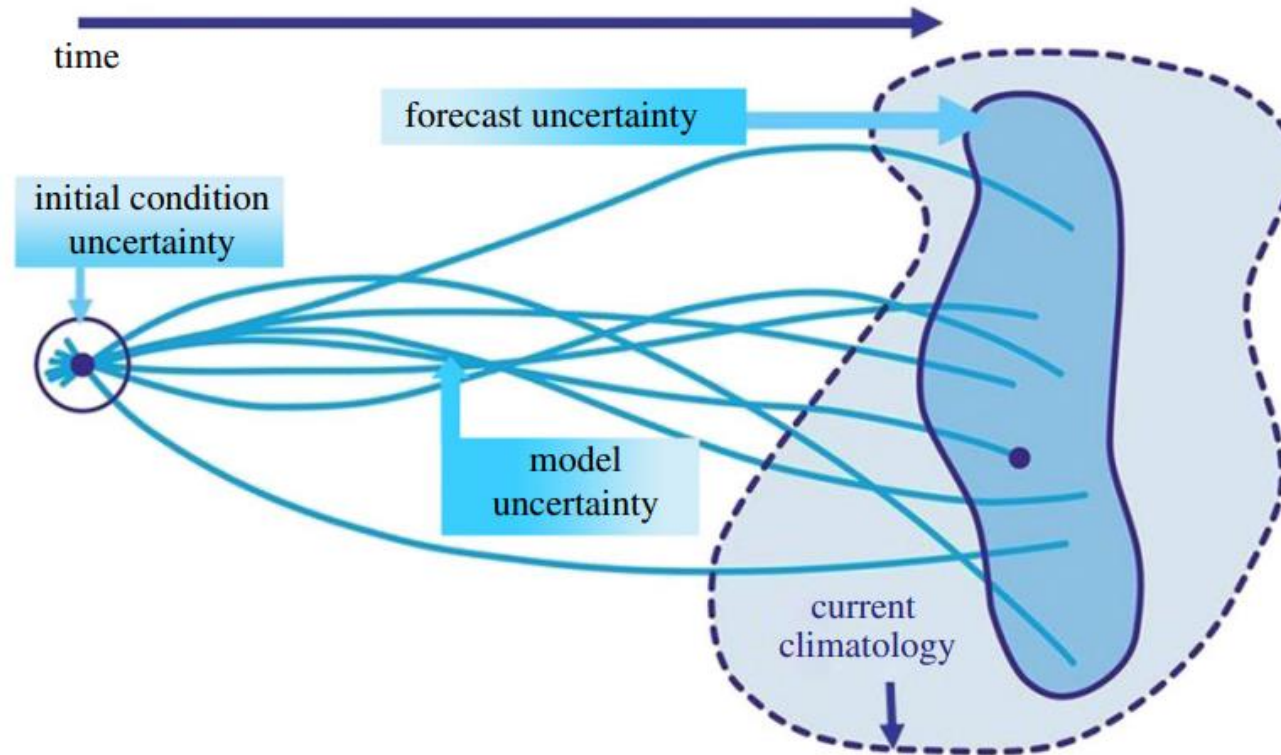
Aggravating Factors

- flood warning (hindered evacuation of waterways)
- observed surge bias of 30 cm
- High coefficients

Uncertainty (.. in models)

Uncertainty: weather and climate prediction

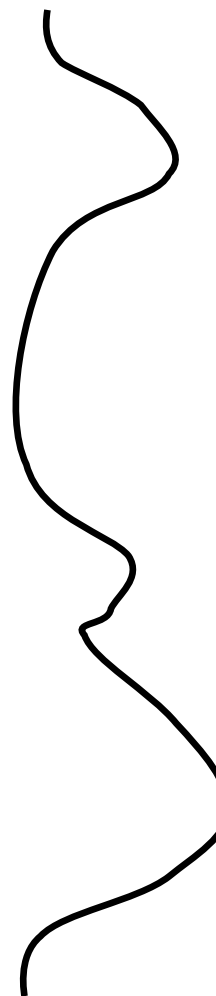
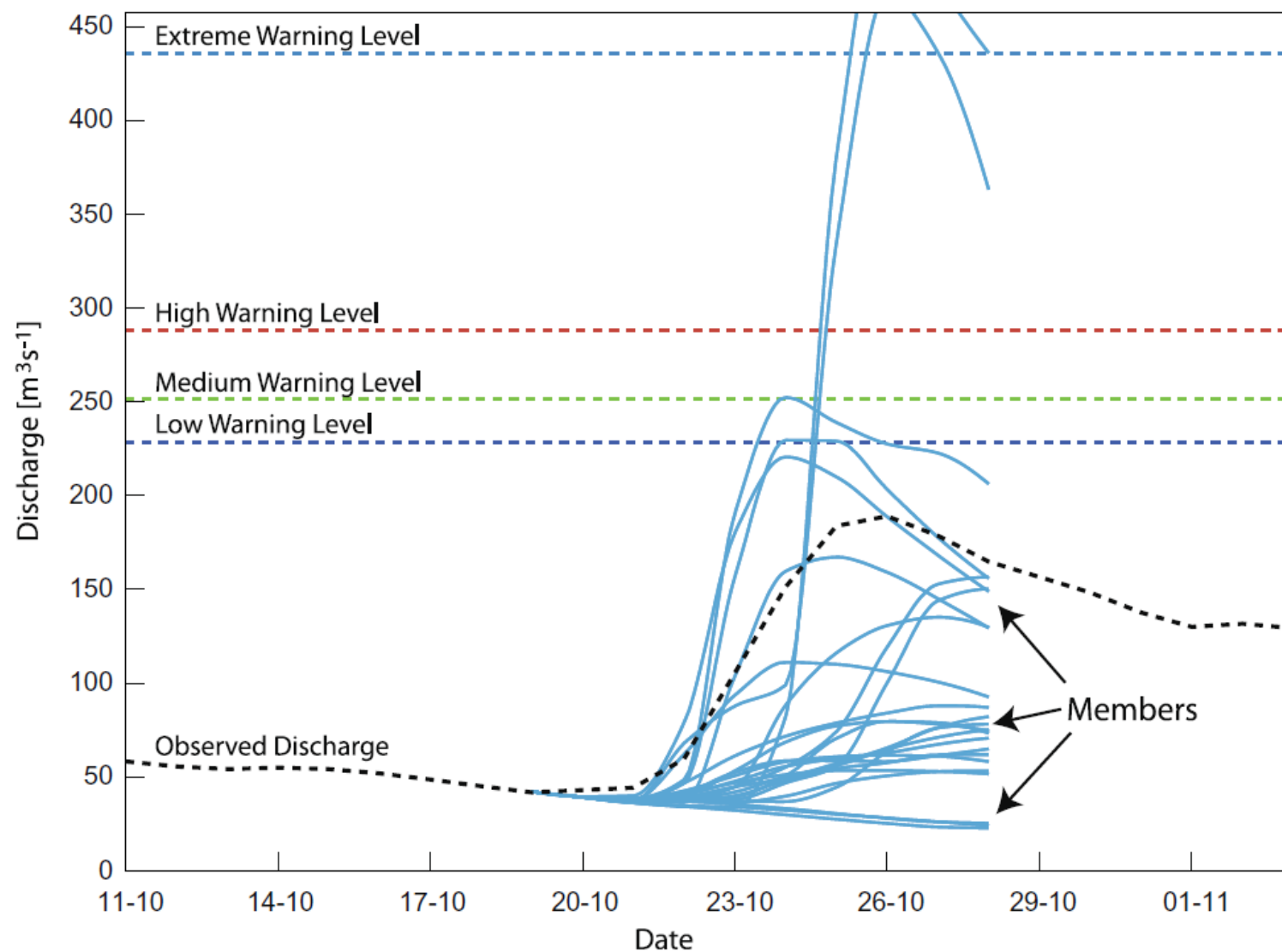
4753



Uncertainty in weather and climate prediction BY JULIA SLINGO,* AND TIM PALMER, Phil. Trans. R. Soc. A (2011) 369, 4751–4767
doi:10.1098/rsta.2011.0161

Figure 2. Schematic of a probabilistic weather forecast using initial condition uncertainties. The blue lines show the trajectories of the individual forecasts that diverge from each other owing to uncertainties in the initial conditions and in the representation of sub-gridscale processes in the model. The dashed, lighter blue envelope represents the range of possible states that the real atmosphere could encompass and the solid, dark blue envelope represents the range of states sampled by the model predictions.

Ensemble Forecast Scenarios & Warnings

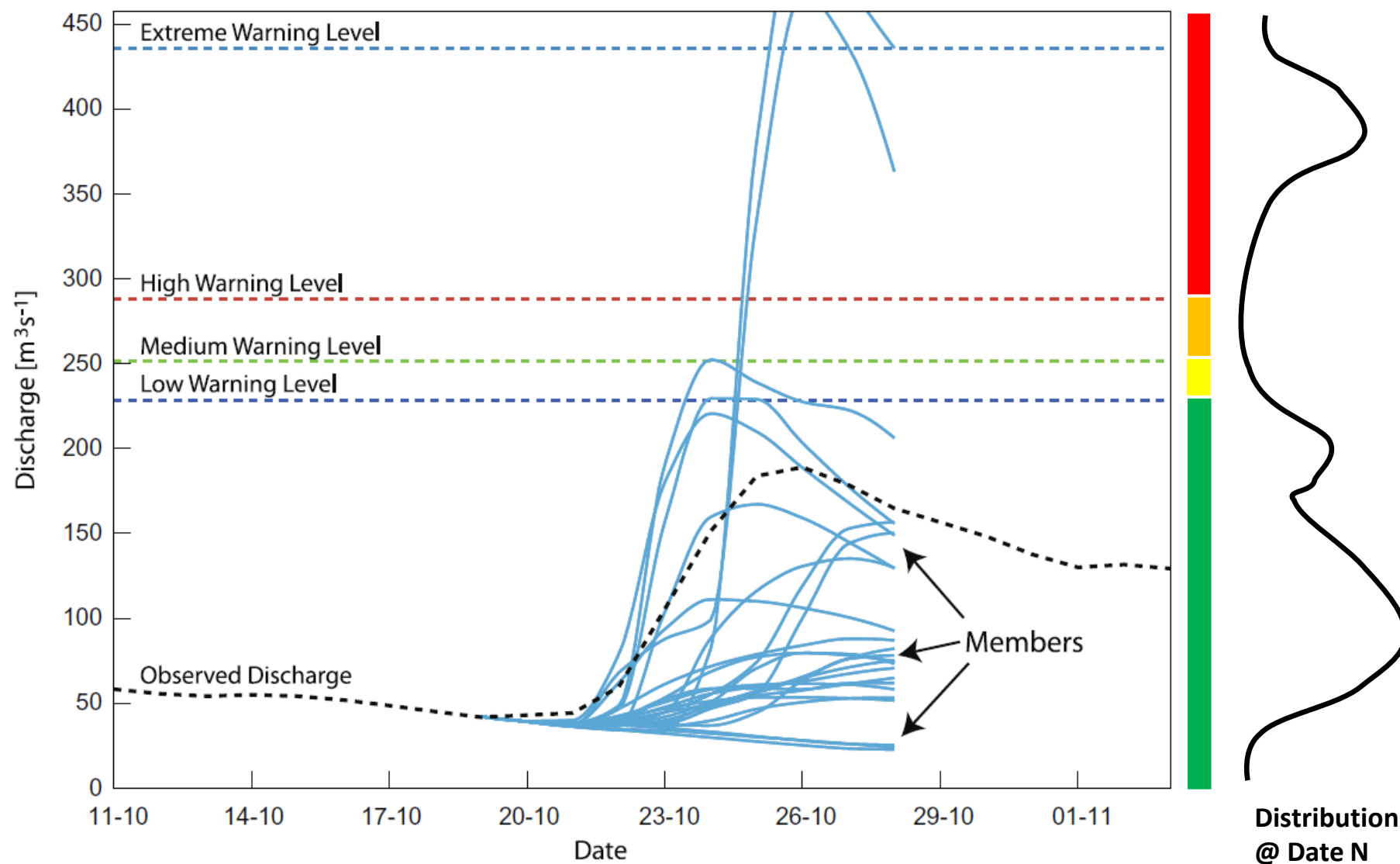


**Distribution
@ Date N**

An example of an ensemble 'spaghetti' hydrograph for a hindcasted flood event. The plot shows the discharge predicted for each ensemble forecast (solid line), the observed discharge (dashed black line) and four flood discharge warning levels (horizontal dashed lines). The example is taken from the October 2007 Flood in Romania on the river Jiu as modelled by the European Flood Alert System (for details of this flooding event see [Pappenberger et al., 2008](#)). Added the orange distribution for additional notes.

Source: H.L. Cloke and F. Pappenberger, 2009, Ensemble flood forecasting: A review, *Journal of Hydrology* 375 (2009) 613–626

Ensemble Forecast Scenarios & Warnings



Une vigilance absolue s'impose des phénomènes dangereux d'intensité exceptionnelle sont prévus...

Soyez très vigilant, des phénomènes dangereux sont prévus ...

Soyez attentif si vous pratiquez des activités sensibles au risque météorologique ...

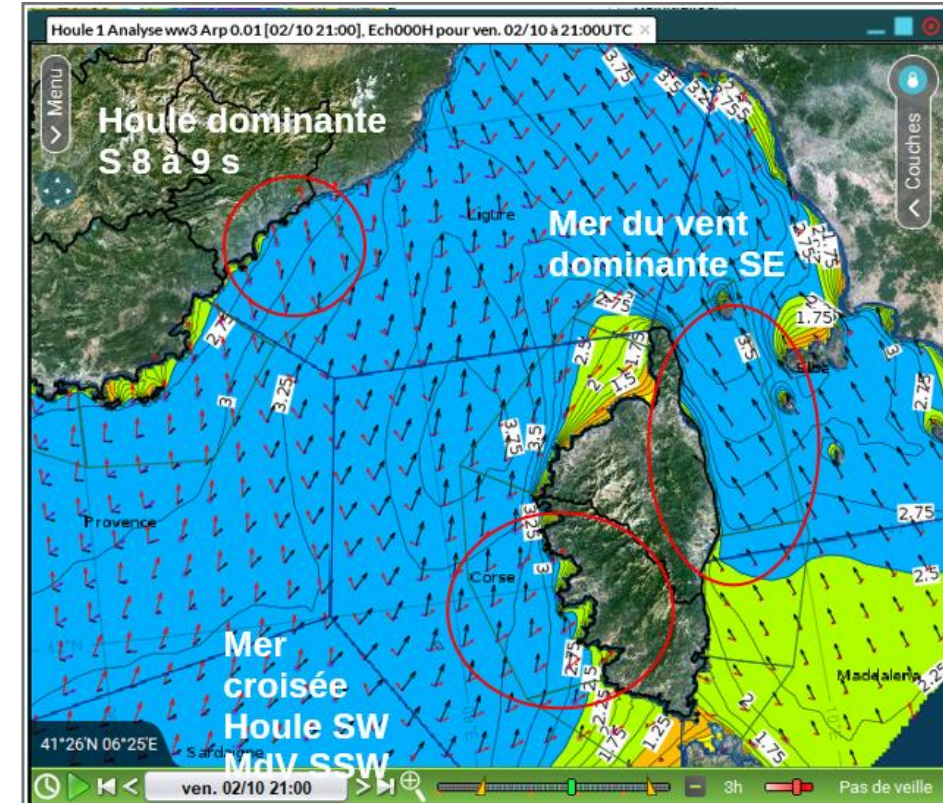
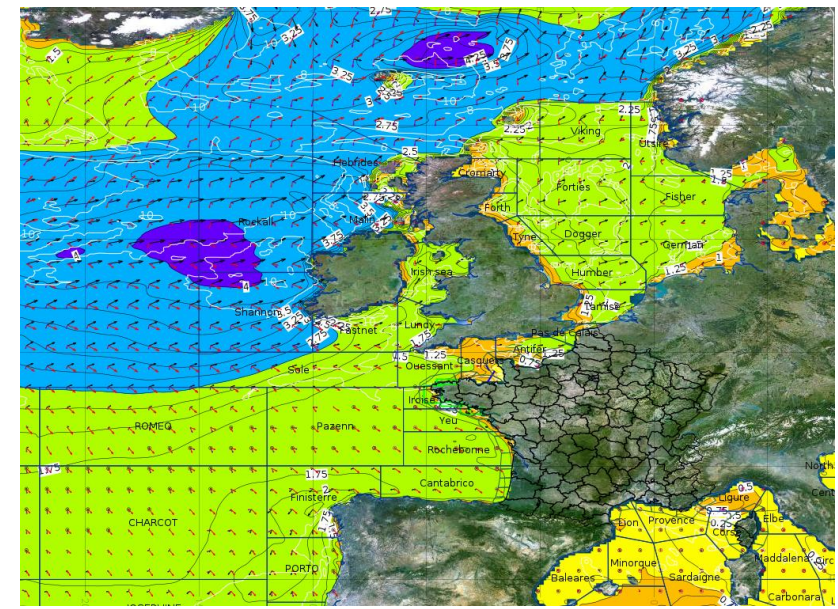
Pas de vigilance particulière.

An example of an ensemble 'spaghetti' hydrograph for a hindcasted flood event. The plot shows the discharge predicted for each ensemble forecast (solid line), the observed discharge (dashed black line) and four flood discharge warning levels (horizontal dashed lines). The example is taken from the October 2007 Flood in Romania on the river Jiu as modelled by the European Flood Alert System (for details of this flooding event see [Pappenberger et al., 2008](#)). Added the orange distribution for additional notes.

Source: H.L. Cloke and F. Pappenberger, 2009, Ensemble flood forecasting: A review, Journal of Hydrology 375 (2009) 613–626

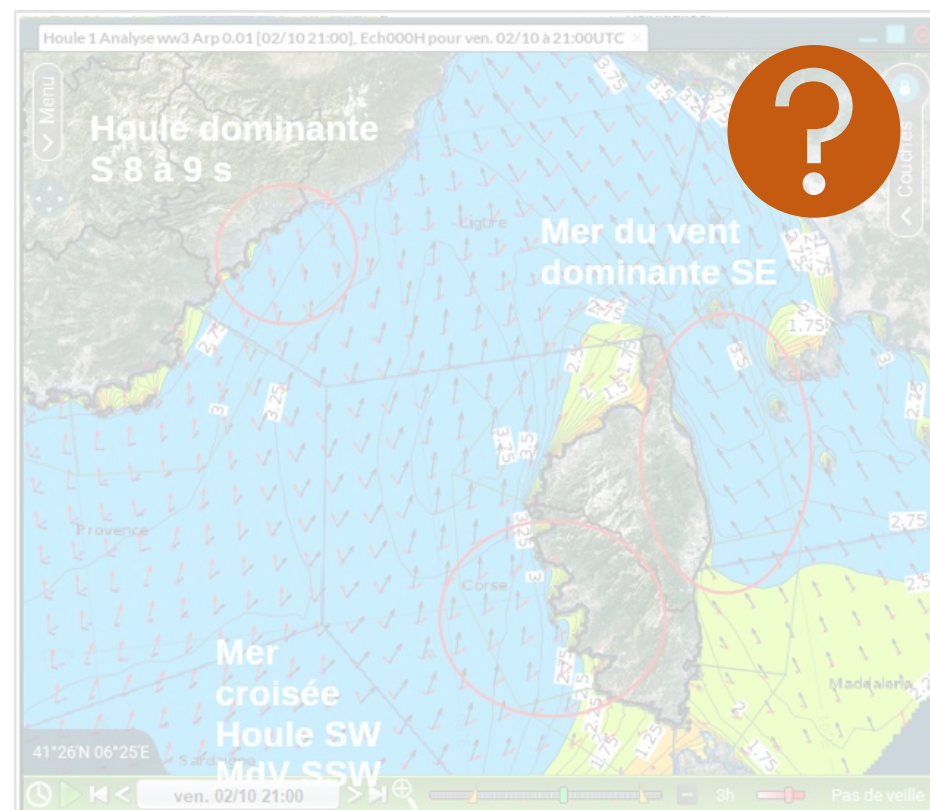
Ensemble Forecast Scenarios

- **34 possible scenarios for each simulation**
- **Simulation Météo France**
 - Surge (Scalar, Spatio-temporal)
 - Wind (Vector, Spatio-temporal)
 - Waves (Vector + scalar, Spatio-temporal)
- **Simulation BRGM**
 - Maximum water height on land (spatial)
 - Maximum sea level reached (spatial)



Ensemble Forecast Scenarios

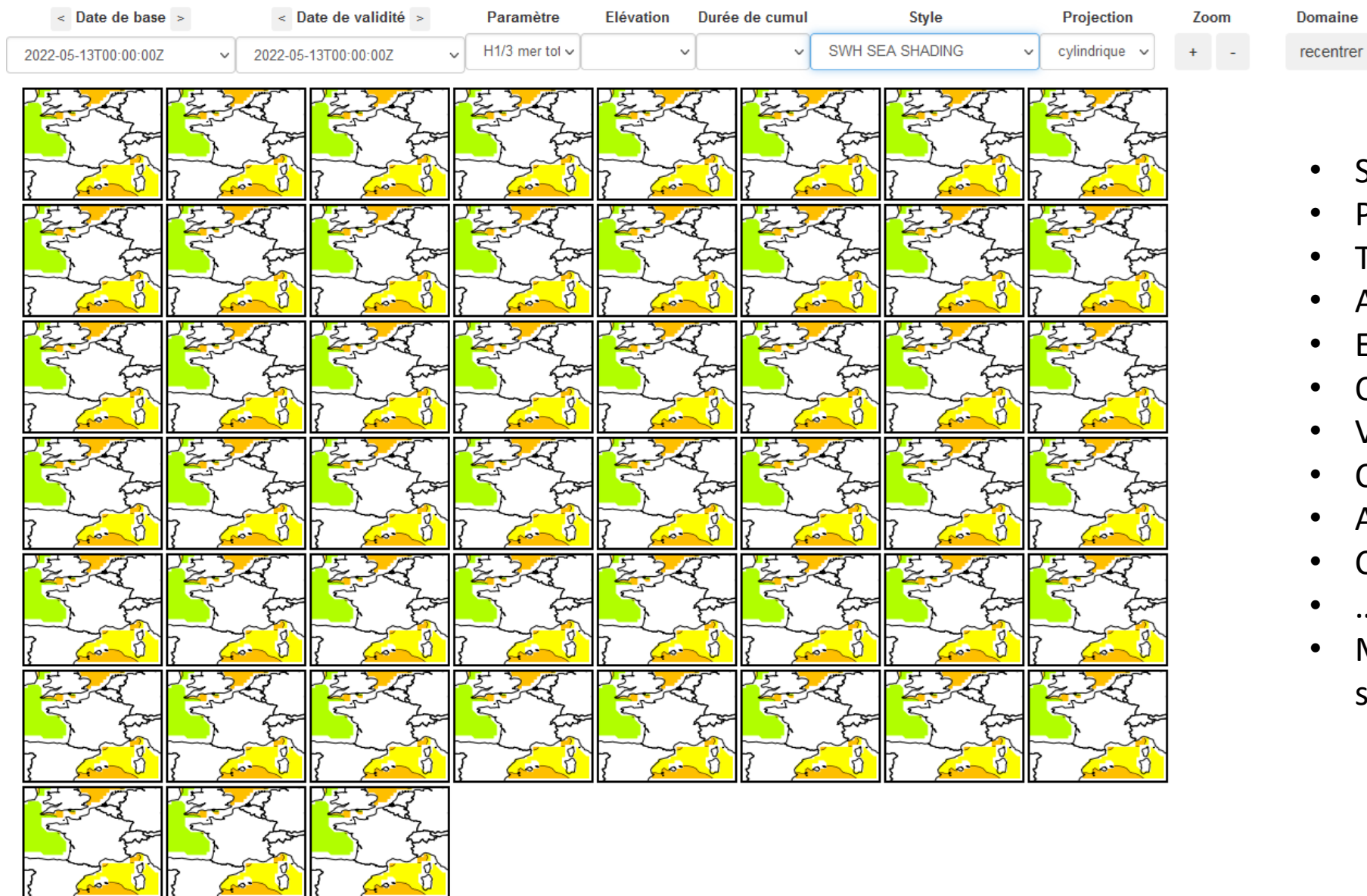
- 34 possible scenarios for each simulation
- Probabilistic Approach
- Simulation Météo France
 - Surge (Scalar, Spatio-temporal) * 34 scenarios
 - Wind (Vector, Spatio-temporal) * 34 scenarios
 - Waves (Vector + scalar, Spatio-temporal) * 34 scenarios
- Simulation BRGM
 - Maximum water height on land (spatial) * 34 scenarios
 - Maximum sea level reached (spatial) * 34 scenarios



Research Focus

- **How to visualize ensemble forecast results to ..**
 - help Météo-France forecasters determine the VVS alert level
 - provide decision support to local authorities
- **How to visualize ..**
 - the main trends in the different scenarios
 - the distribution of results according to the different scenarios
 - the level of uncertainty

Mode Multistamp - Vignettes



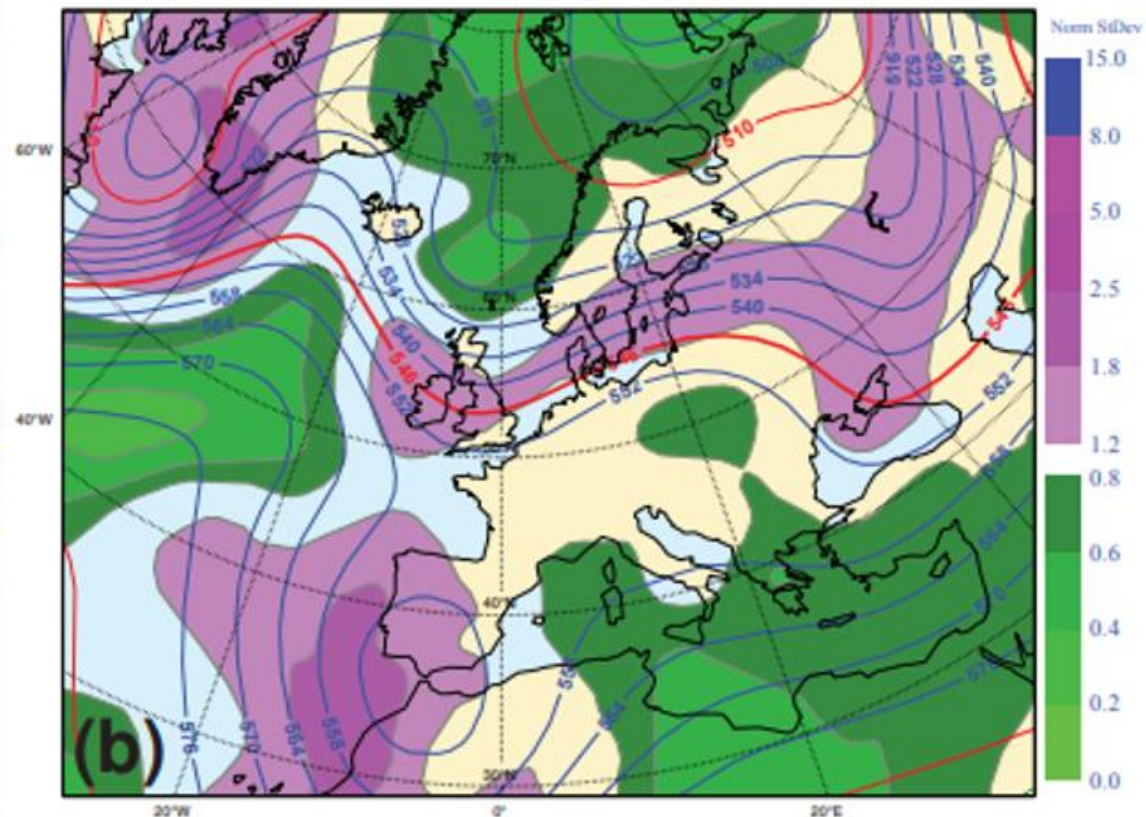
- Scenarios
- Patterns
- Trends
- Anomalies
- Extremes
- Outliers
- Variability
- Consistency
- Agreement
- Confidence
- ...
- Most probable situation?

ni: Sat,21MAR2015 06Z Val: Fri,27MAR2015 06Z

500 hPa Geopotential (Isohypsens: 516 552 576 gpdam)



Thursday 19 March 2015 00UTC ECMWF Forecast t+120 VT: Tuesday 24 March 2015 00UTC
500hPa Geopotential Ensemble Mean and Normalised Standard Deviation (shaded)



Ensemble forecast products.

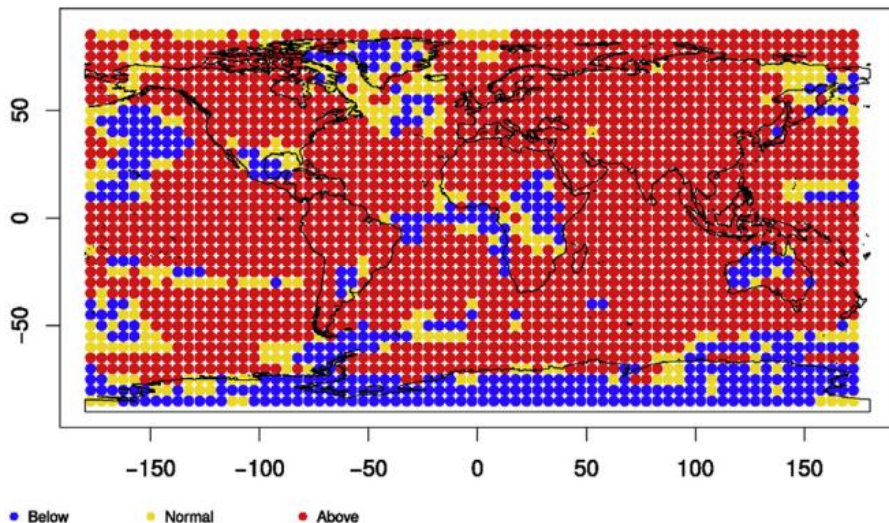
(a) “**Spaghetti plot**”, comparing contour lines of three isovalues of the 500 hPa geopotential height field of an NCEP GEFS forecast at 144 lead time. (Courtesy of G. Muller, www.wetterzentrale.de.) ”

(a) ECMWF forecast product showing **ensemble mean** of 500 hPa geopotential height (contour lines) and normalized standard deviation of the same field (filled contours). (Courtesy of ECMWF.)

Visualization in Meteorology – A Survey of Techniques and Tools for Data Analysis Tasks
Marc Rautenhaus, Michael Bottinger, Stephan Siemen, Robert Hoffman, “
Robert M. Kirby, Mahsa Mirzargar, Niklas Rober, and Rüdiger Westermann
Visualisation and computer graphics

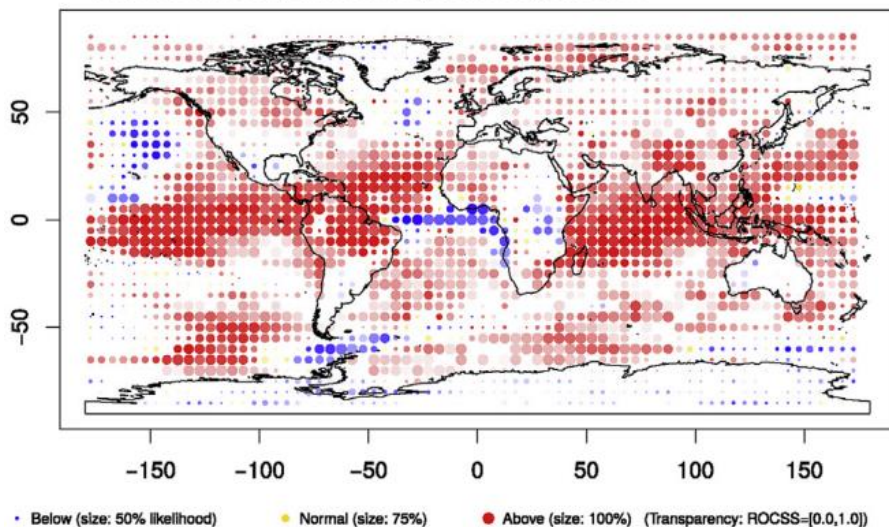
a. 2-meter air temperature, Dec to Feb, 2016

Reference data: NCEP; Hindcast: CFS (24 members); 1983–2010



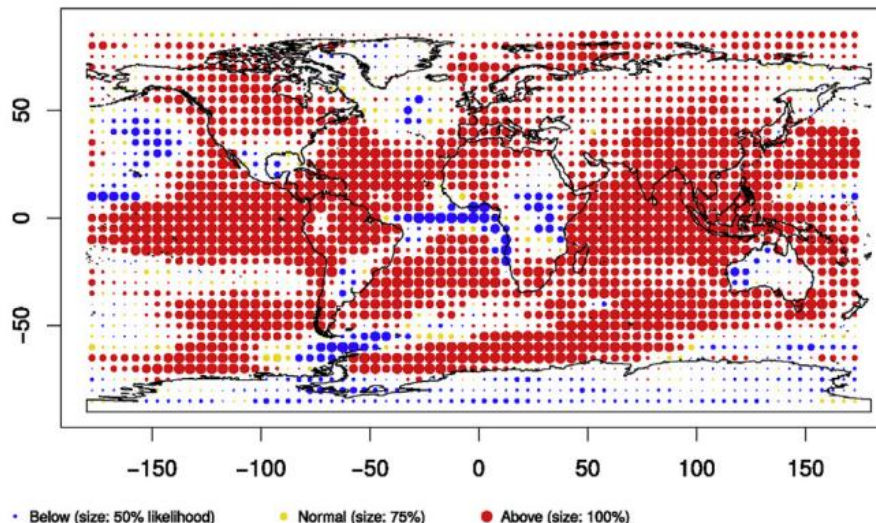
c. 2-meter air temperature, Dec to Feb, 2016

Reference data: NCEP; Hindcast: CFS (24 members); 1983–2010



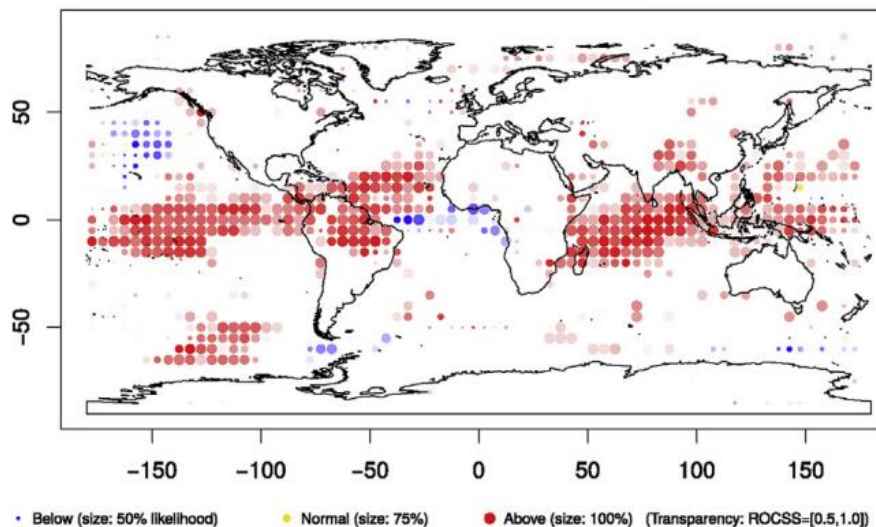
b. 2-meter air temperature, Dec to Feb, 2016

Reference data: NCEP; Hindcast: CFS (24 members); 1983–2010



d. 2-meter air temperature, Dec to Feb, 2016

Reference data: NCEP; Hindcast: CFS (24 members); 1983–2010

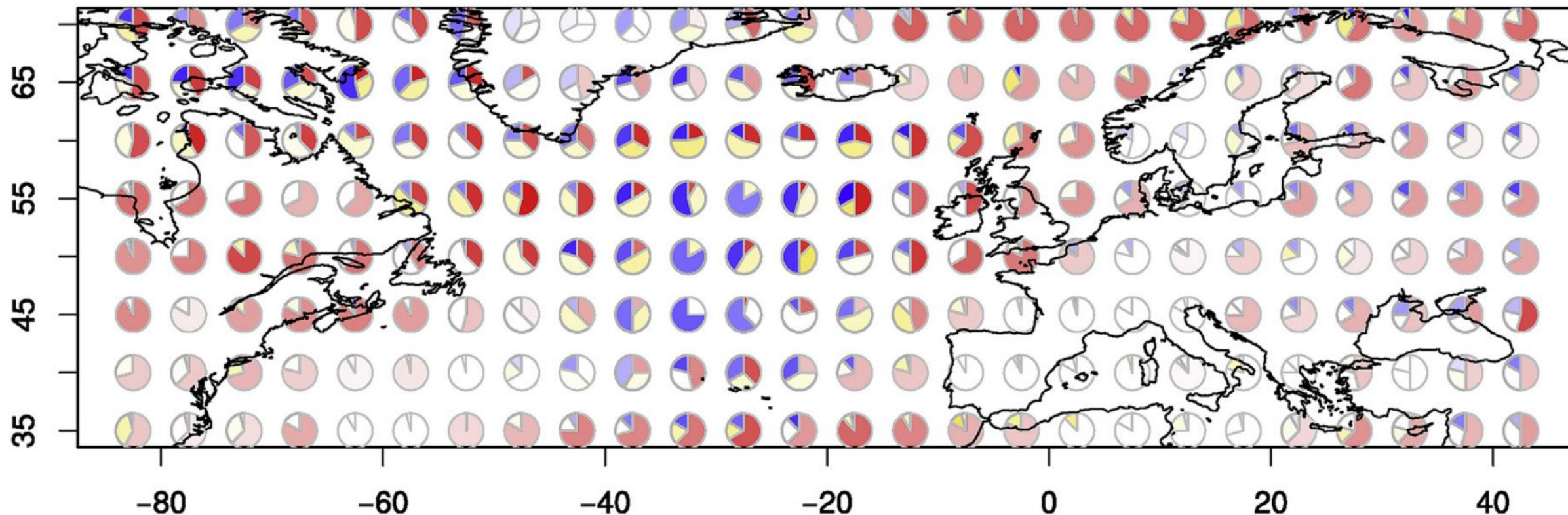


- Bubble plot** for the December, 2015 to February, 2016 2-m air temperature forecast issued in November, 2015 by the CFSv2 forecast system (24 members). The **colour of the bubbles** indicates the most likely tercile.
- As a), but representing the probability of the most likely tercile by the size of the bubble. **Bubble size** for different forecast probabilities are represented in the legend for reference.
- As b), but including ROCSS value as **transparency**. Forecasts where past skill is worse than the climatology (negative ROCSS) are whitened out. NCEP Reanalysis data is considered as reference to compute the skill score.
- As c), but **transparency range** only covers ROCSS values from 0.5 to 1 as indicated in the legend

An R package to visualize and communicate uncertainty in seasonal climate prediction M.D. Frías a, *, M. Iturbide b, R. Manzanar b, J. Bedia a, J. Fernandez a, S. Herrera a, A.S. Cofino~ a, J.M. Gutierrez b, Environmental Modelling & Software 99 (2018) 101e110

2-meter air temperature, Dec to Feb, 2016

Reference data: NCEP; Hindcast: CFS (24 members); 1983–2010

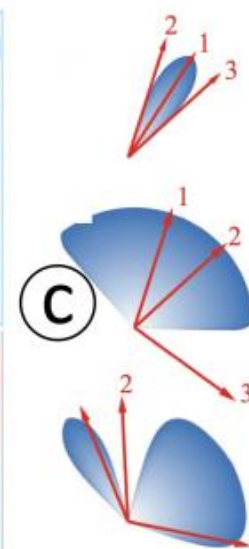
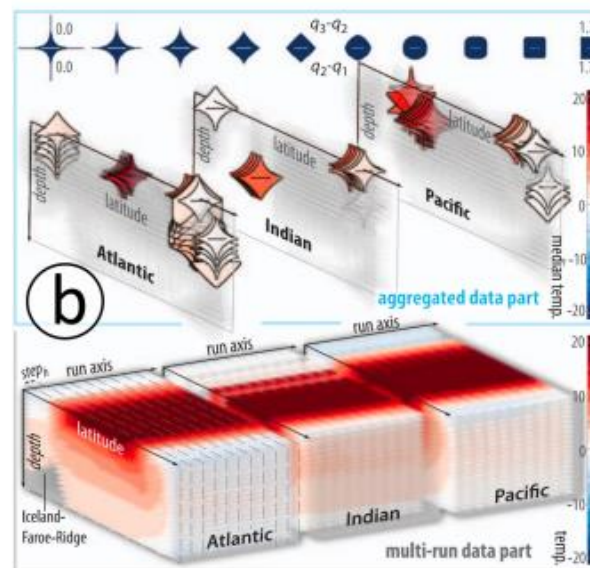
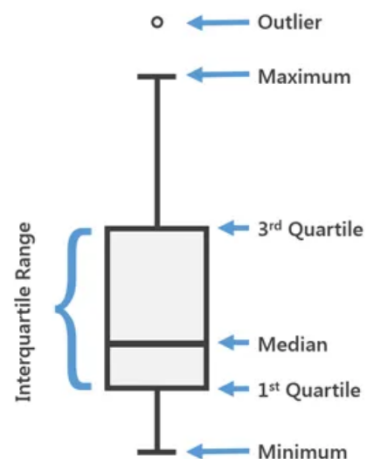


As Fig. 1c but including the probability forecast for each tercile **as sectors of a 3-piece pie chart** in a zoom over the North Atlantic region. Transparency shows the skill score of the probabilistic system for each tercile. Only positive values are plotted.

An R package to visualize and communicate uncertainty in seasonal climate prediction M.D. Frías a, *, M. Iturbide b, R. Manzananas b, J. Bedia a, J. Fernandez a, S. Herrera a, A.S. Cofino~ a, J.M. Gutierrez b, Environmental Modelling & Software 99 (2018) 101e110

Boxplot

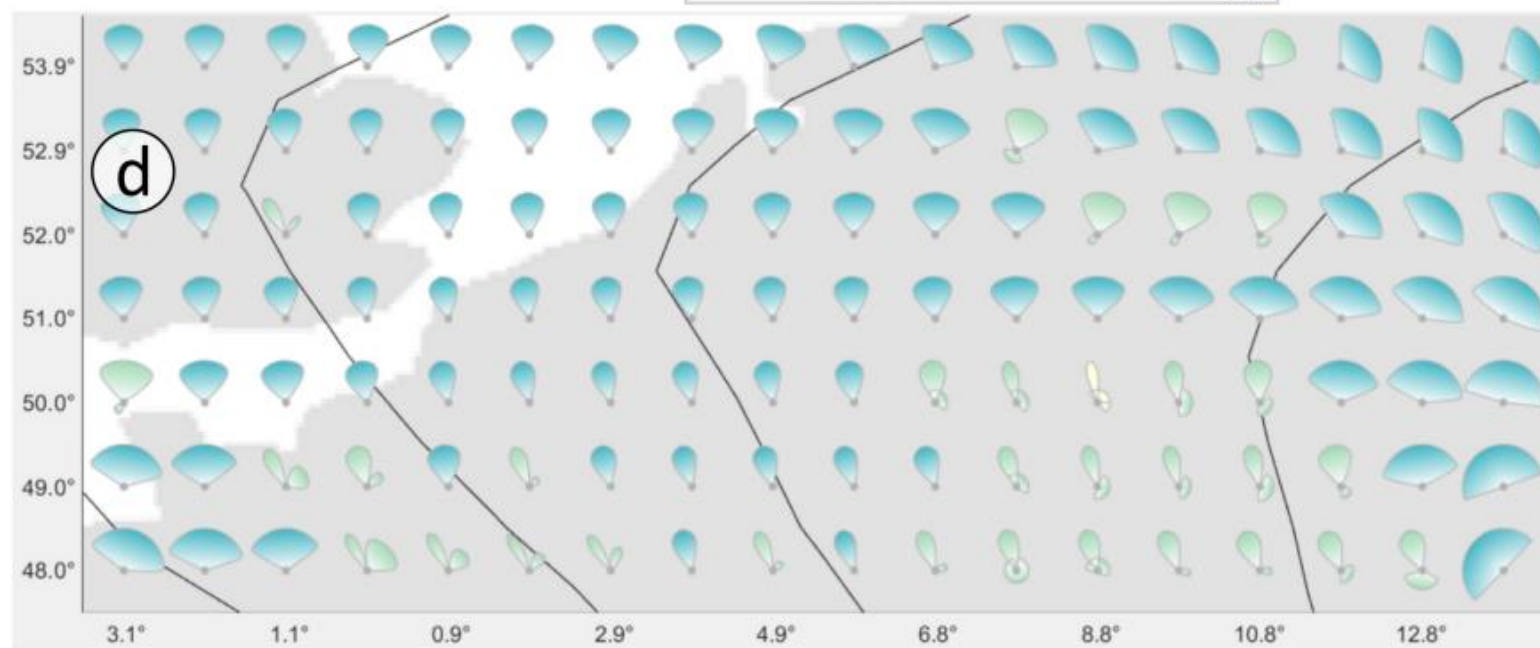
(a)



POINT BASED METHODS

Aggregation before visualization:

- Boxplot elements. [by Vamshi Krishna Dande | Medium](#)
- The color, size, upper/lower shape of **superellipse glyphs** encode the median, interquartile range, and subquartile range of all members, image courtesy of Kehrer et al. [62] © 2011 IEEE.
- Using **lobular glyphs** to show the modality of vector directions at each grid point;
- results of placing **lobular glyphs** across the simulation field. Image courtesy of Jarema et al. [63] © 2015 IEEE.



Visualization and Visual Analysis of Ensemble Data: A Survey Junpeng Wang, Subhashis Hazarika, Cheng Li, and Han-Wei Shen, Member, IEEE, Visualisation and computer graphics

[a\) Boxplot: Explain Percentiles , Quartiles , IQR\(Inter Quartile Range\) and Box Plot | by Vamshi Krishna Dande | Medium](#)

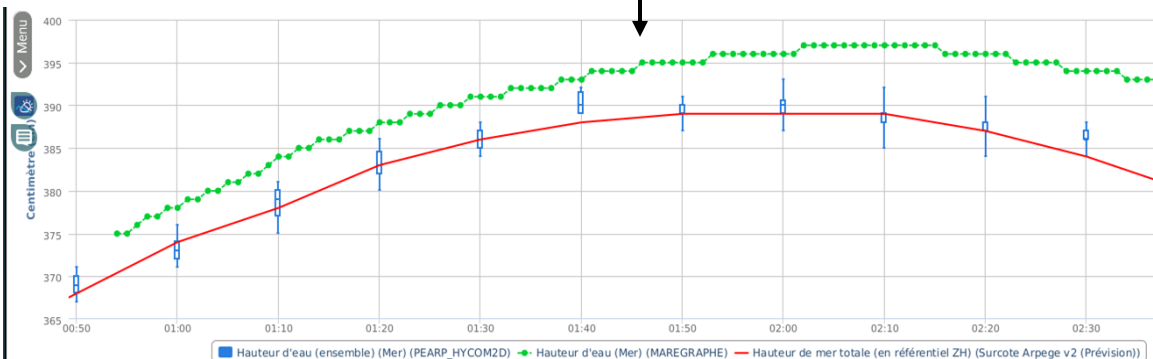
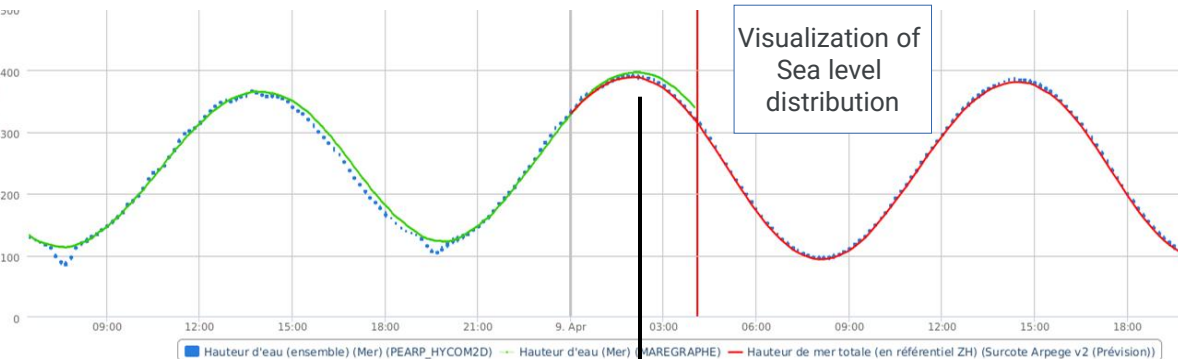
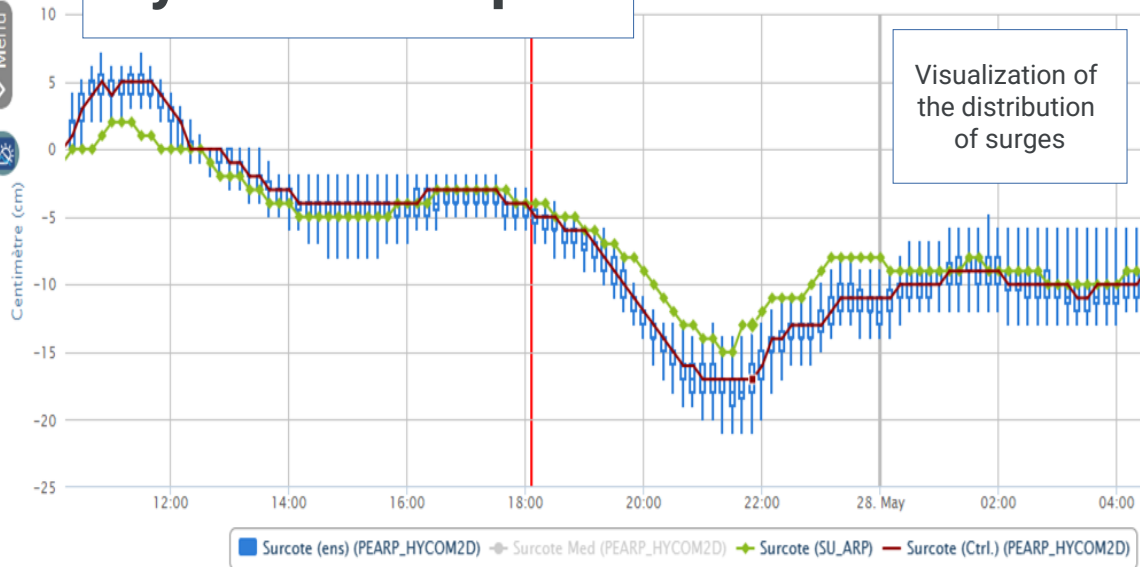
By reference port

2 – PE of Surges



Visualization of the distribution of surges

Visualization of Sea level distribution



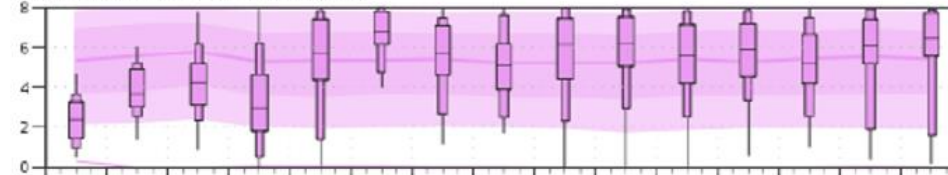
ENS Meteogram

Reading. United Kingdom 51.39°N 0.83°W (EPS land point) 51 m

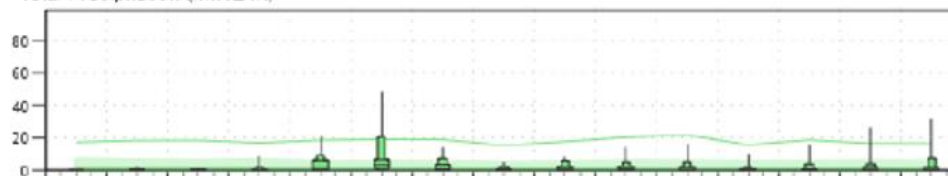
Extended Range Forecast based on ENS distribution Sunday 7 June 2015 00



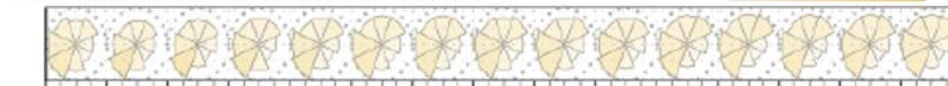
Daily mean of Total Cloud Cover (okta)



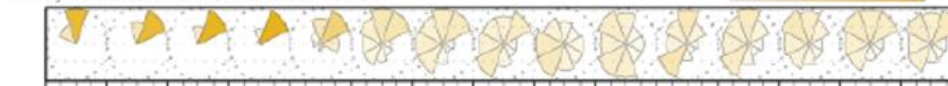
Total Precipitation (mm/24h)



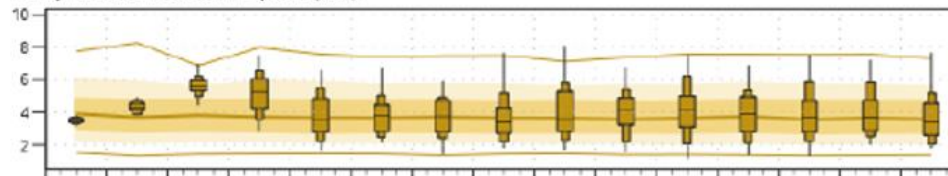
M-Climate of the distribution of 10m Wind Direction



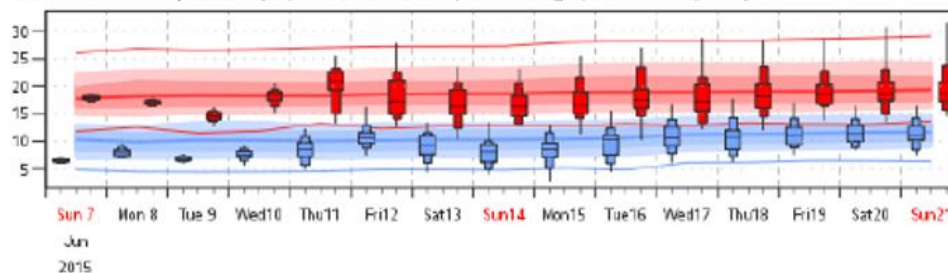
Daily Distribution of 10m Wind Direction



Daily mean of 10m Wind Speed (m/s)



2m min/max Temperature (°C) reduced to 51 m (station height) from 96 m (T319)



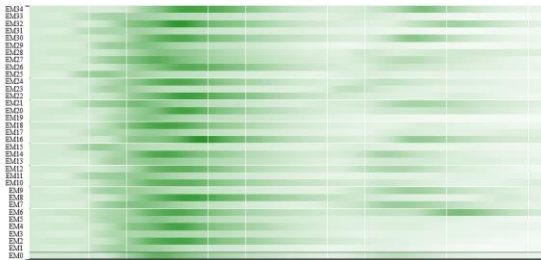
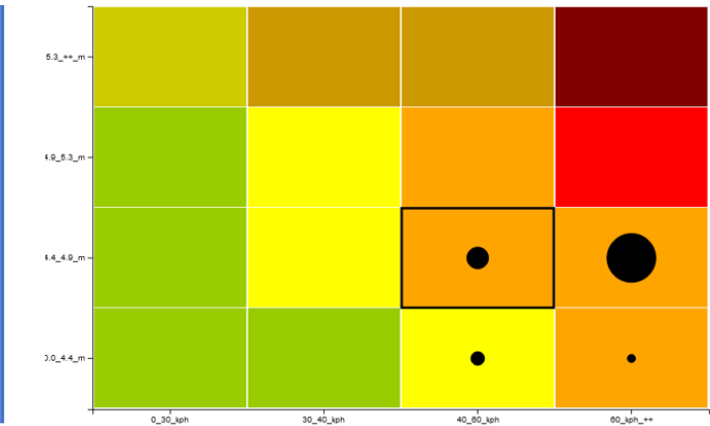
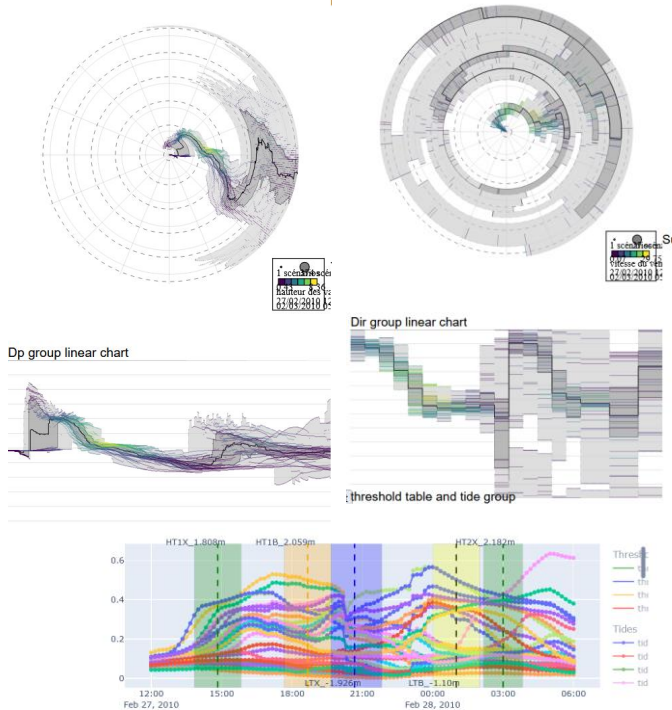
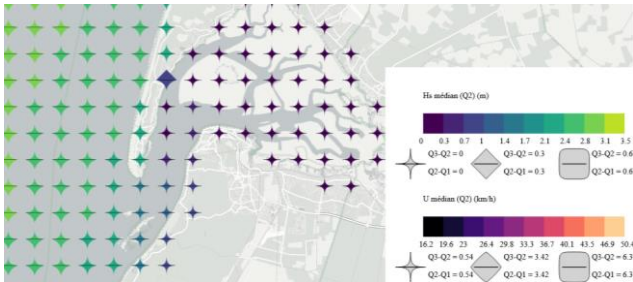
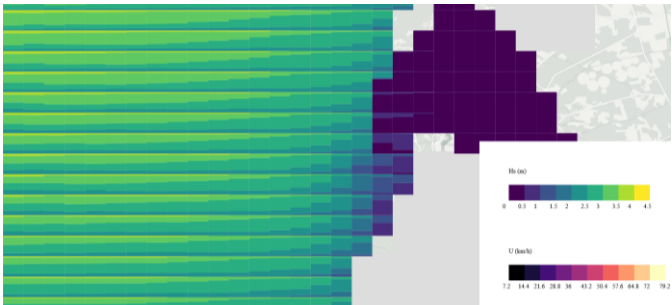
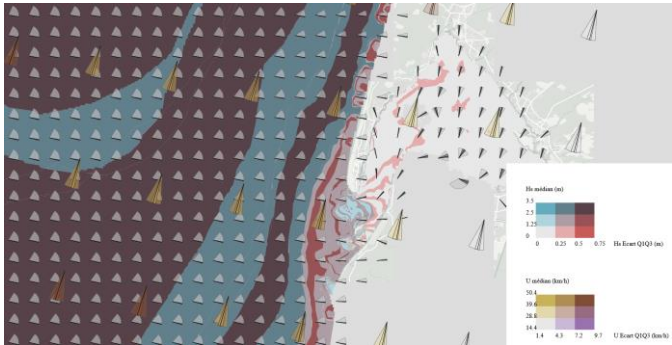
M-Climate:
99%
90%
75%
median
25%
10%
1%

max
90%
75%
median
25%
10%
min

M-Climate: this stands for Model Climate. It is a function of lead time, date (+/-15days), and model version. It is derived by rerunning a 11 member ensemble over the last 20 years twice a week (1980 realisations). M-Climate is always from the same model version as the displayed ENS data.

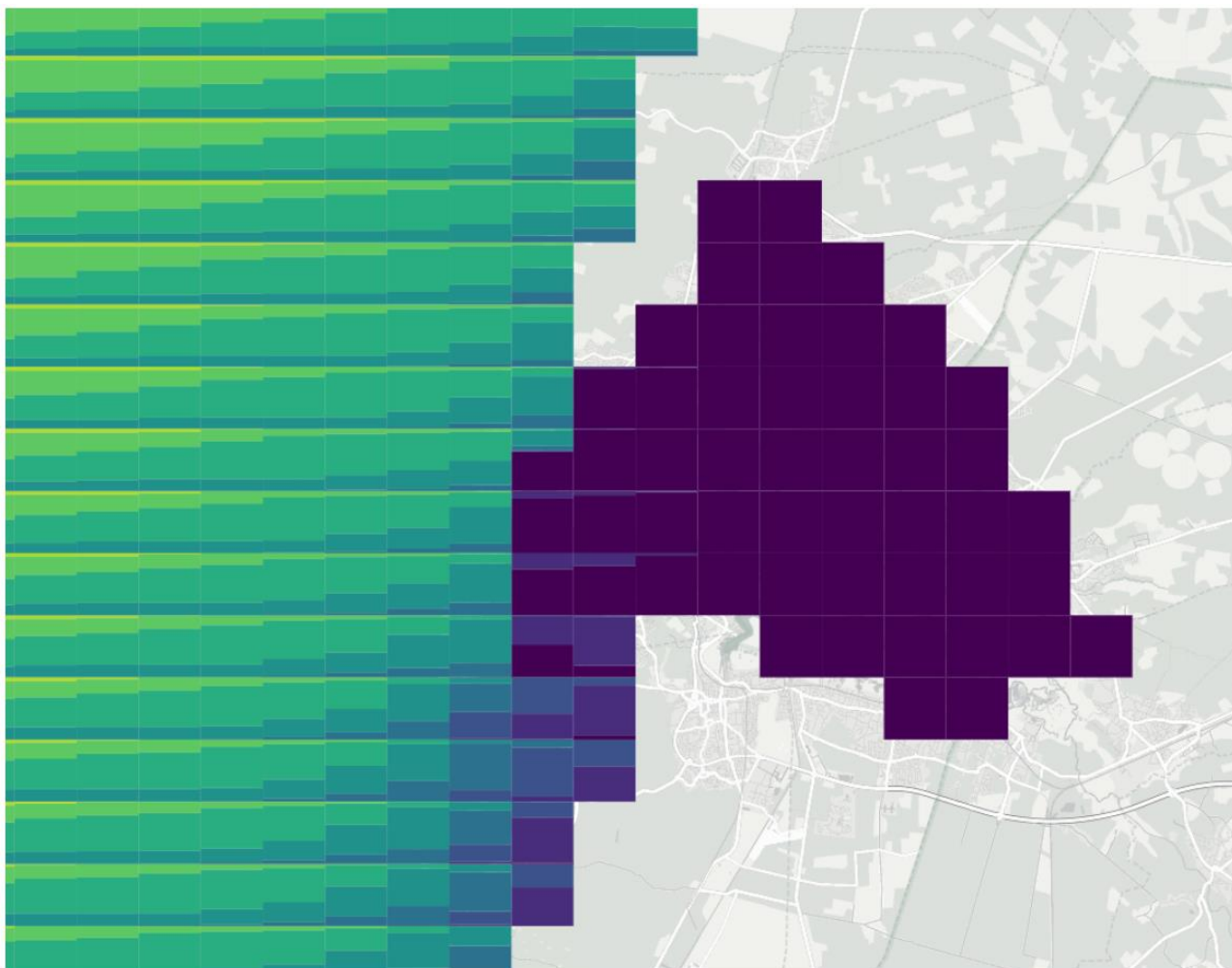
Visualization of confidence in weather forecasts: ECMWF's users, Designing information to focus on what is relevant, Anna Ghelli, ECMWF Anna.ghelli@ecmwf.int

Visual analysis strategy of meteo oceanic conditions from Meteo France outputs



Wave Height (Hs) - Grid

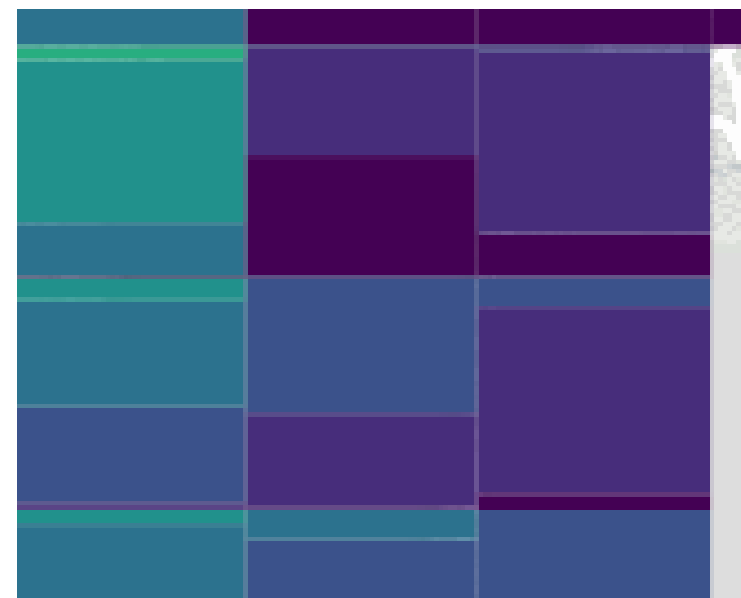
Detailed view of the distribution of wave height (Hs) values



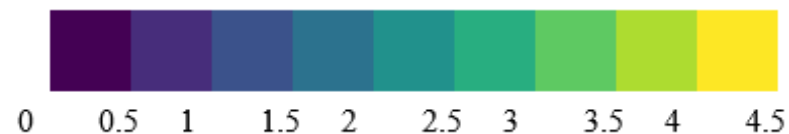
<https://eurostat.github.io/gridviz/>

Grid representation with glyphs

representing the distribution of wave height (Hs) values according to different classes by a color scale



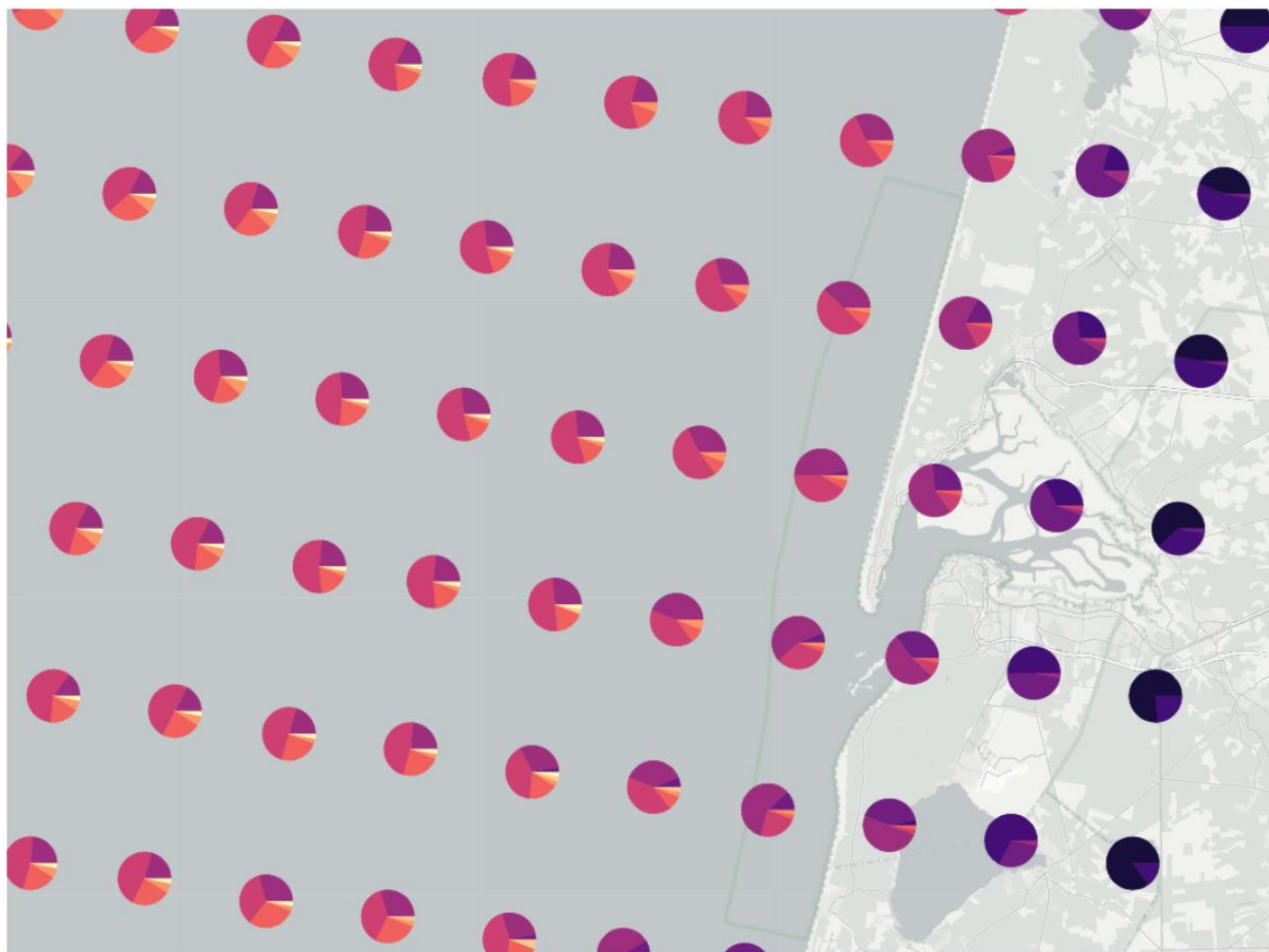
Hs (m)



- Ability to change layer visibility
- Multi-scale

Wind Speed (U) - Point

Detailed view of the distribution of wind speed (U) values

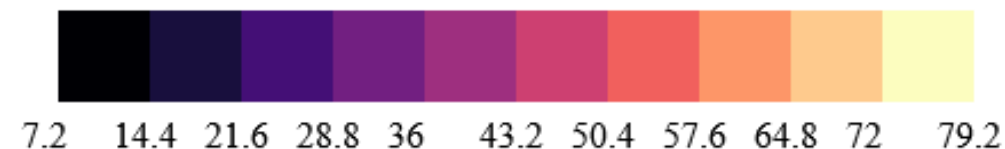


Point representation with glyphs

representing the distribution of wind speed values (U) according to different classes by a color scale



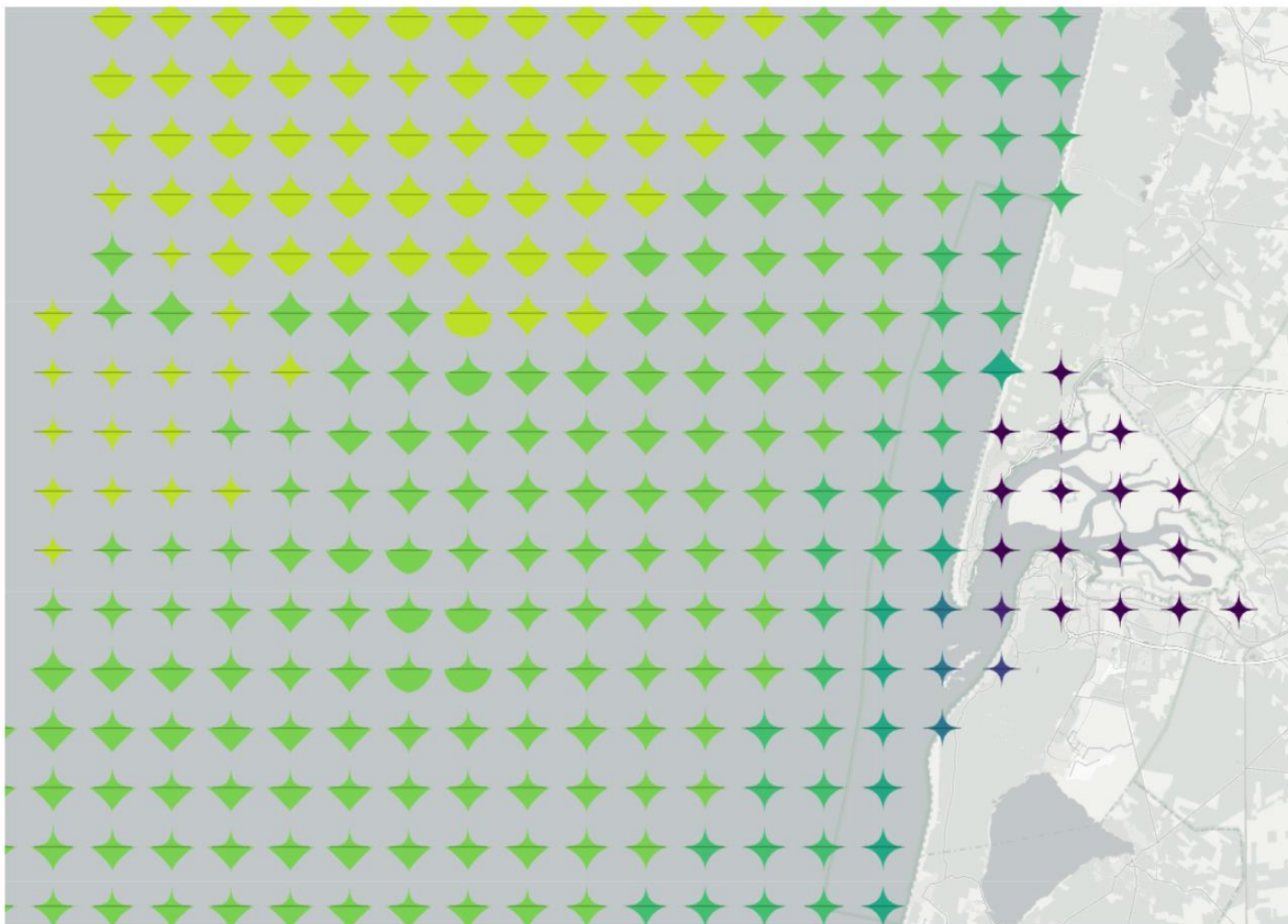
U (km/h)



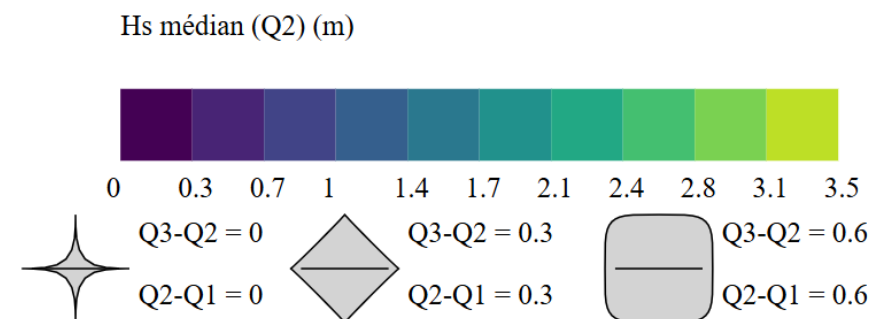
- Ability to change layer visibility
- Multi-scale

Wave Height (Hs) - Grid

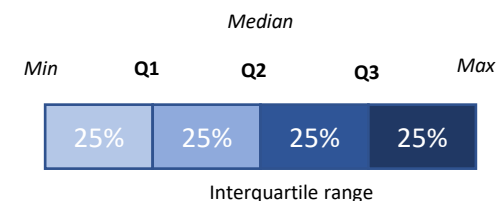
Detailed view of the distribution of wave height (Hs) values



Grid representation with glyphs
representing the statistical distribution of wave height values (Hs)

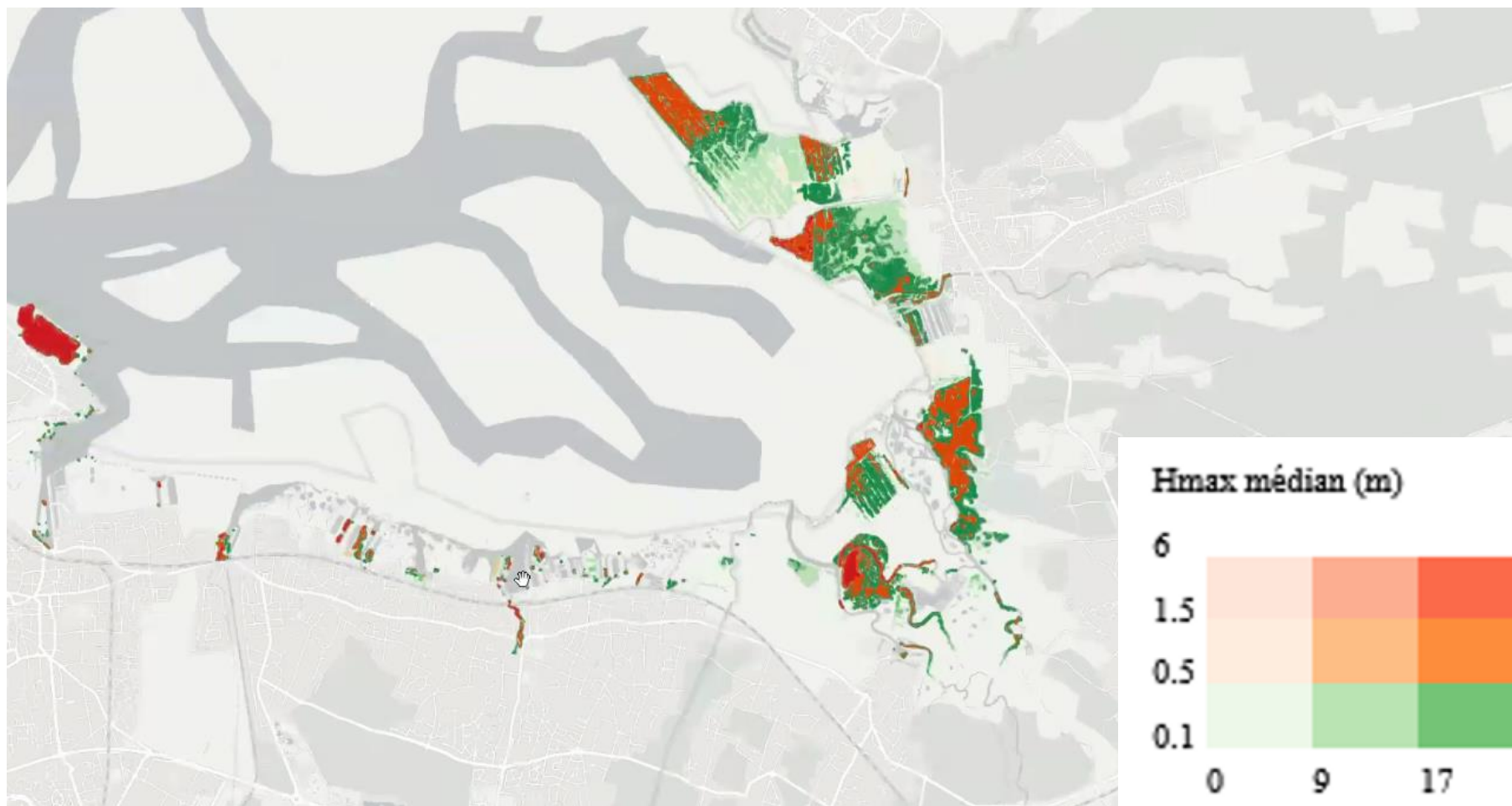


- **Median value:** color scale
- **Interquartile range:** glyph size
- **1st quartile-median range (Q2-Q1):** shape and size of the lower part of the glyph
- **3rd quartile-median range (Q3-Q2):** shape and size of the upper part of the glyph

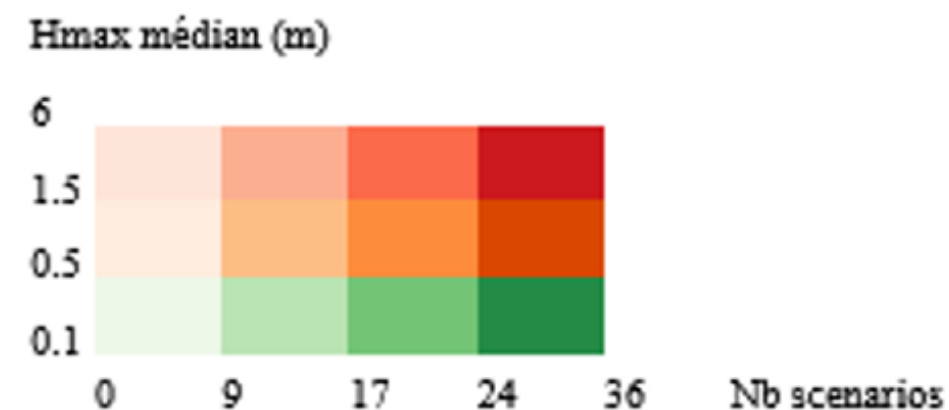


- Ability to change layer visibility
- Multi-scale

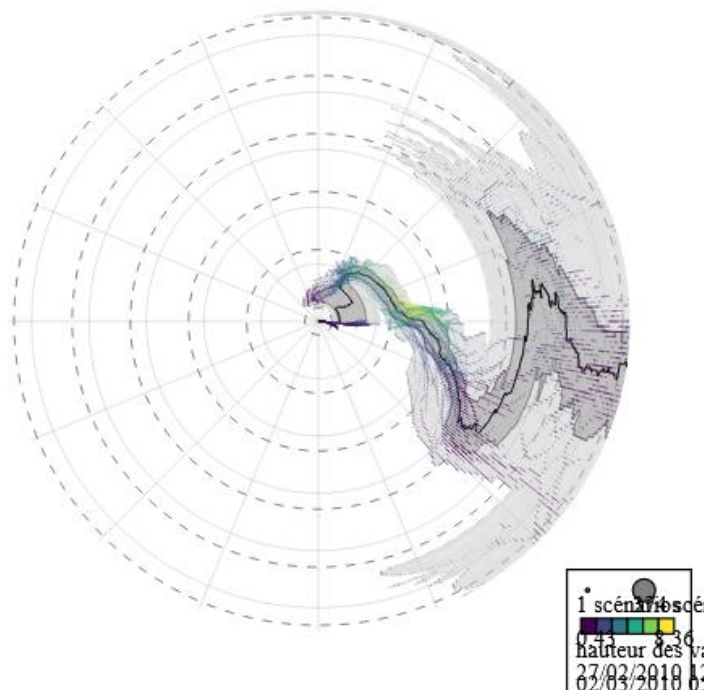
Visualise results of simulations from BRGM



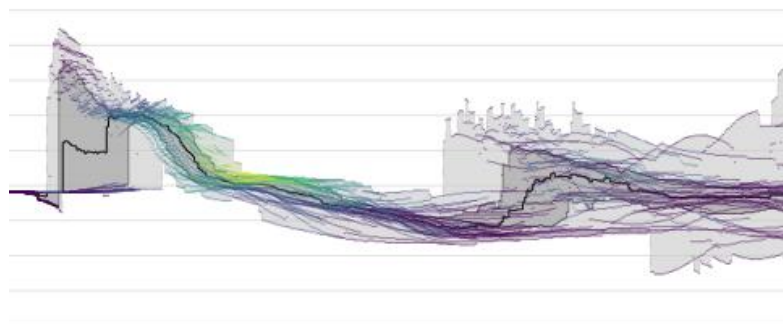
Vector polygon
representation of water
level > 0,1 meters,
bivariate colors, values
are medians



Multivariate analysis @ 1 reference point



Dp group linear chart



Wave/wind direction vs. time plots

1 wind rose plot

- Radius: time
(ex: centre 27/10, border 29/10)
- Angle: direction

1 line diagram

- X: time
- Y: direction

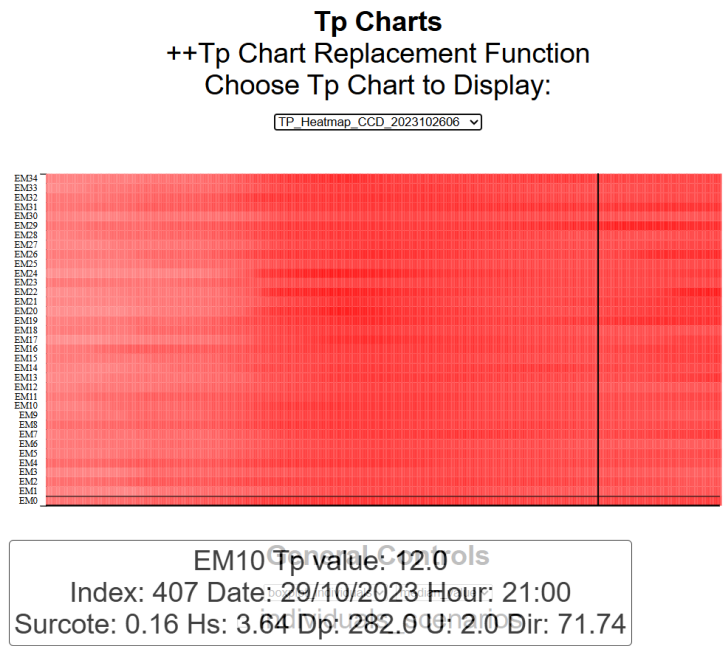
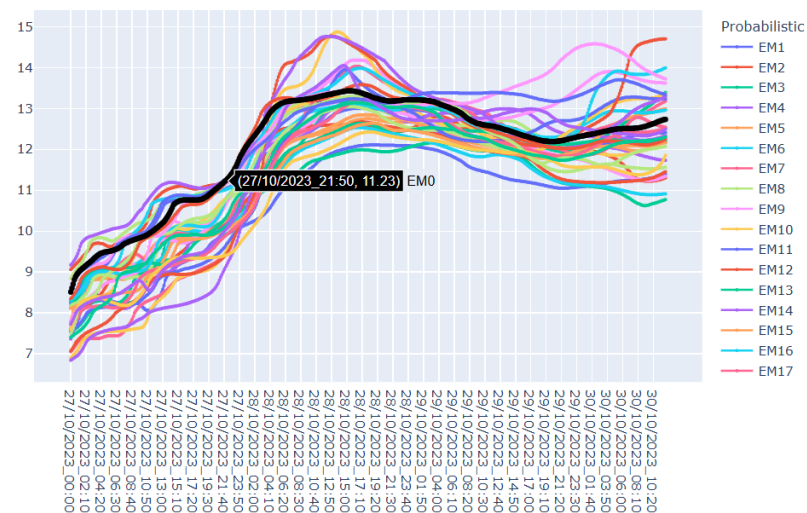
Median value: black line

Interquartile range: gray envelope

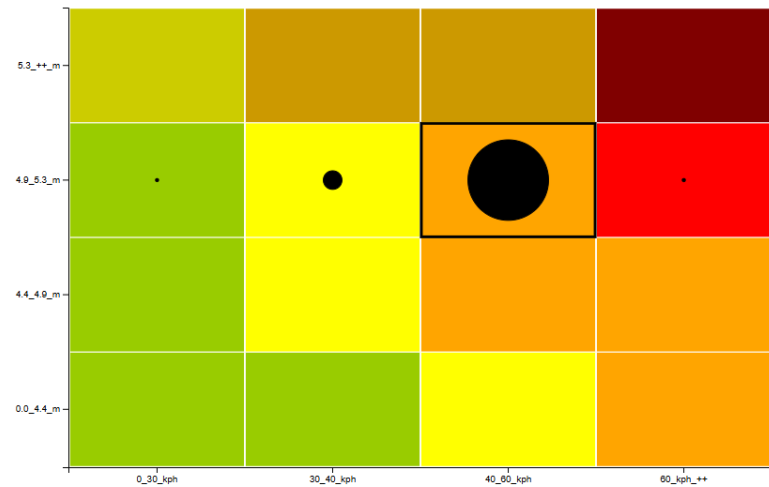
Min-max range: light gray envelope

Synchronization of the two diagrams Time zoom
2*2 diagrams (wind and waves)

Visualisations Proposed/Tested @ specific location



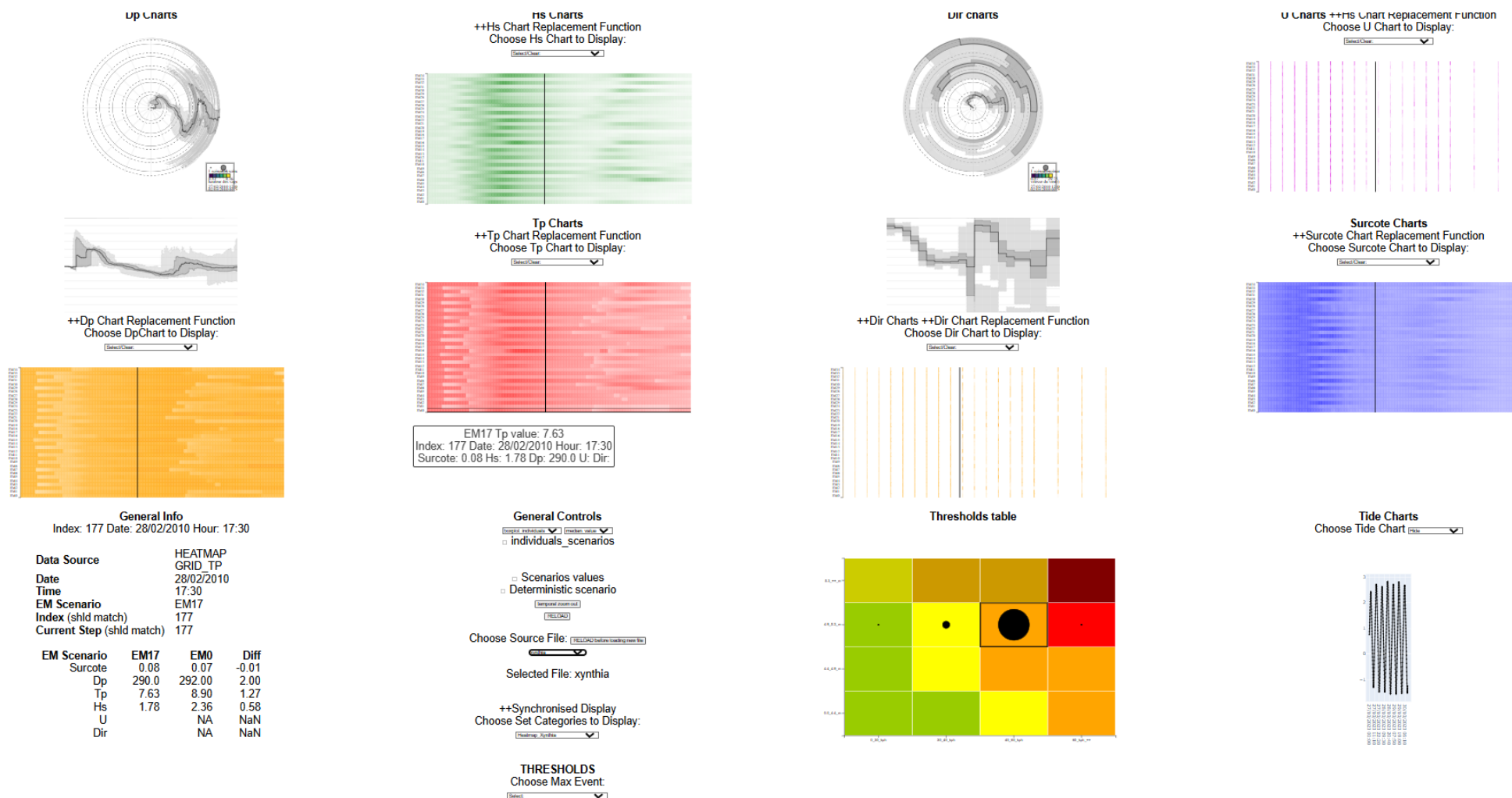
Thresholds table



Number of deterministic scenarios in this cell: **0**.
 Number of probabilistic scenarios in this cell: **6**.
 The Probabilistic Scenarios in this cell are: **E2,E3,E4,E6,E8,E34**

Matrix divided according to: sea level * wind speed
Proportional symbols: number of scenarios falling into a cell of the matrix
Black box: cell in which the deterministic scenario falls.

Visualisations Proposed/Tested/UI (Browser)



Some forecaster feedback

- The visualisations **received good feedback** in general.
- Some visualisations are difficult to interpret (bivariate color scale).
- Some are redundant.
- The threshold table visualisation provides a good overview and starting point for more detailed analysis.
- Some require more time for analysis and therefore not fit for time constrained operational workflows.
- The interactive linked views facilitate data exploration and comparison.

Merci beaucoup!